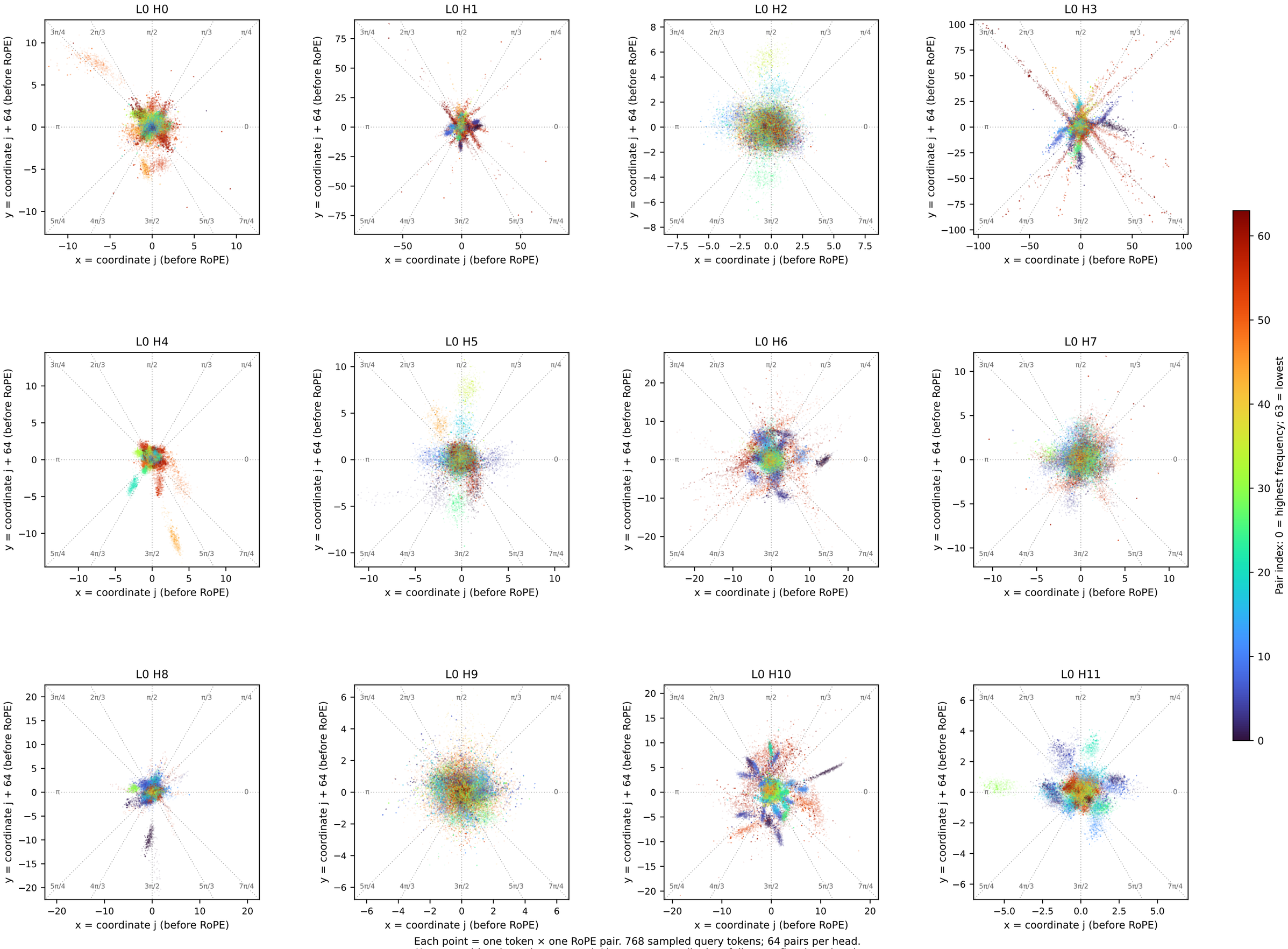
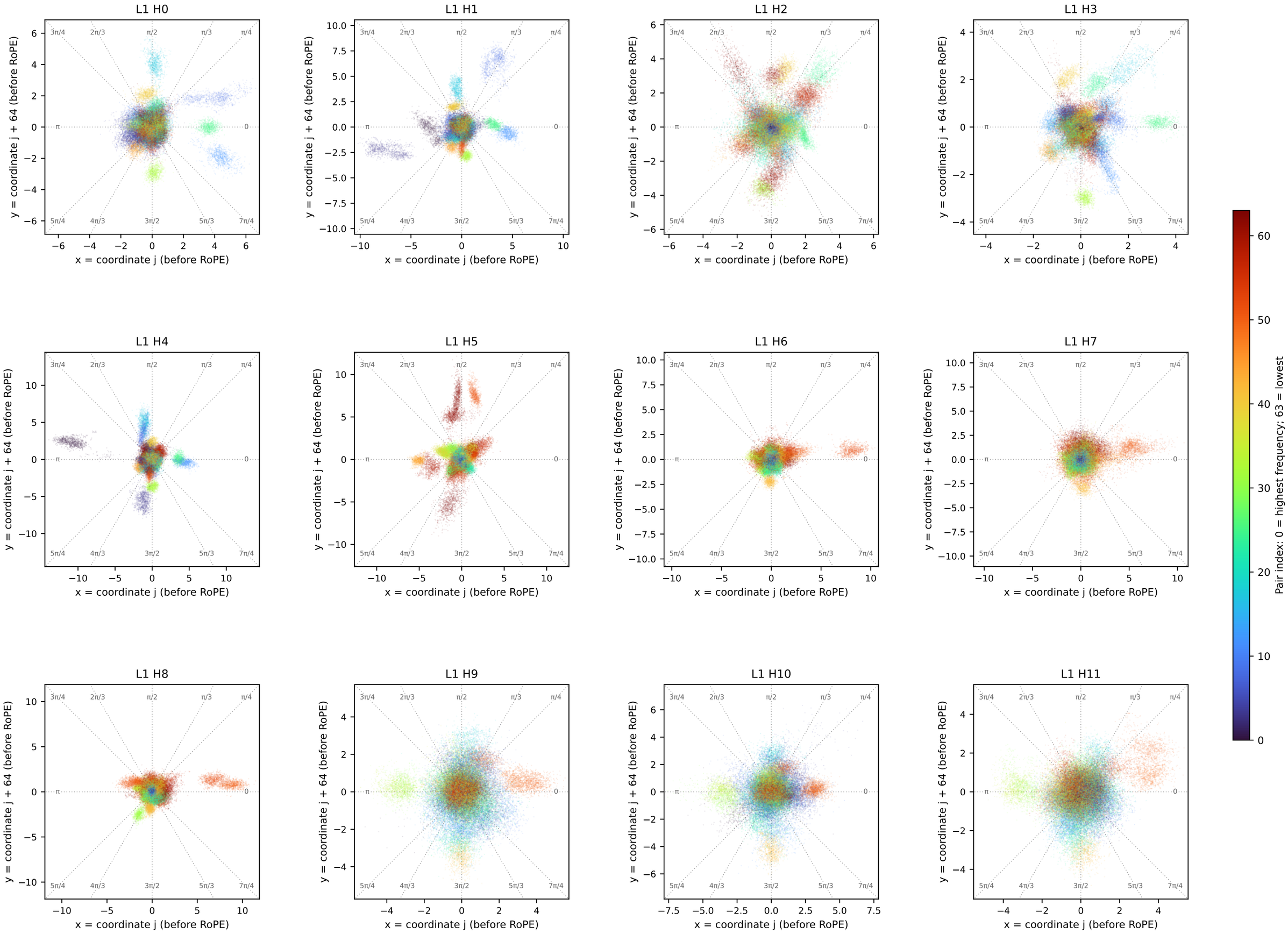
Qwen queries · layer 0 · axes fitted separately for each head



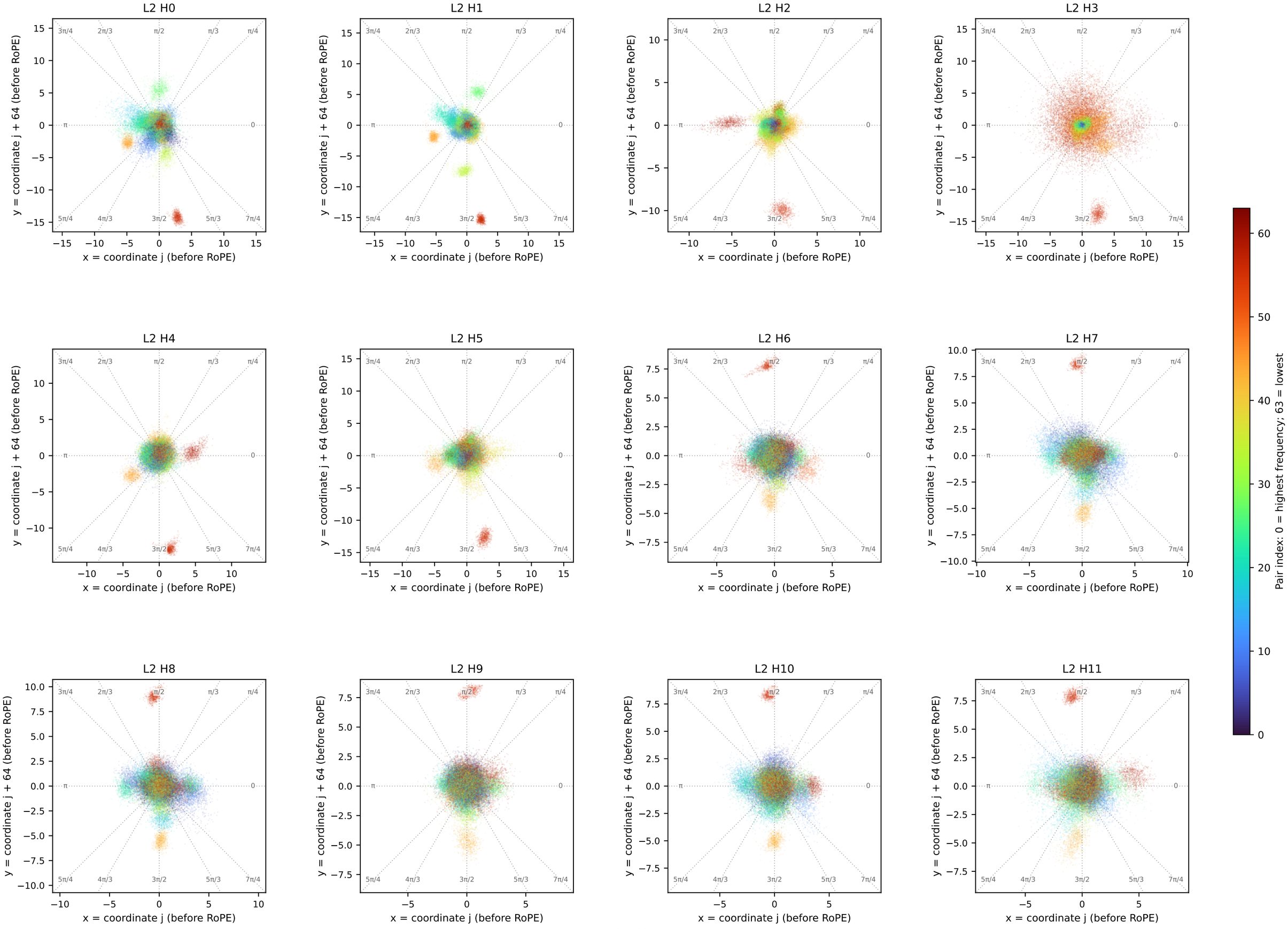
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 1 · axes fitted separately for each head



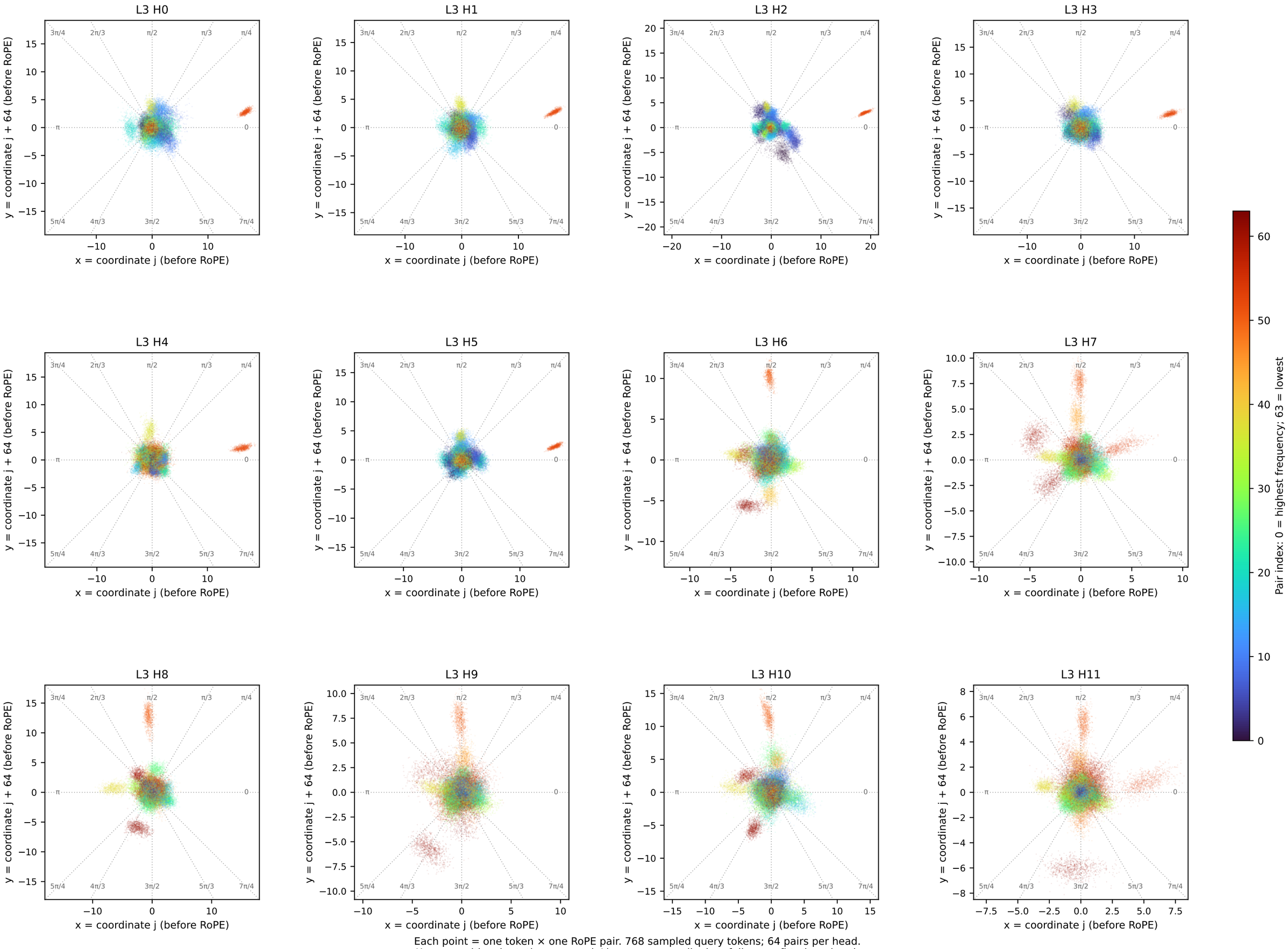
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 2 · axes fitted separately for each head



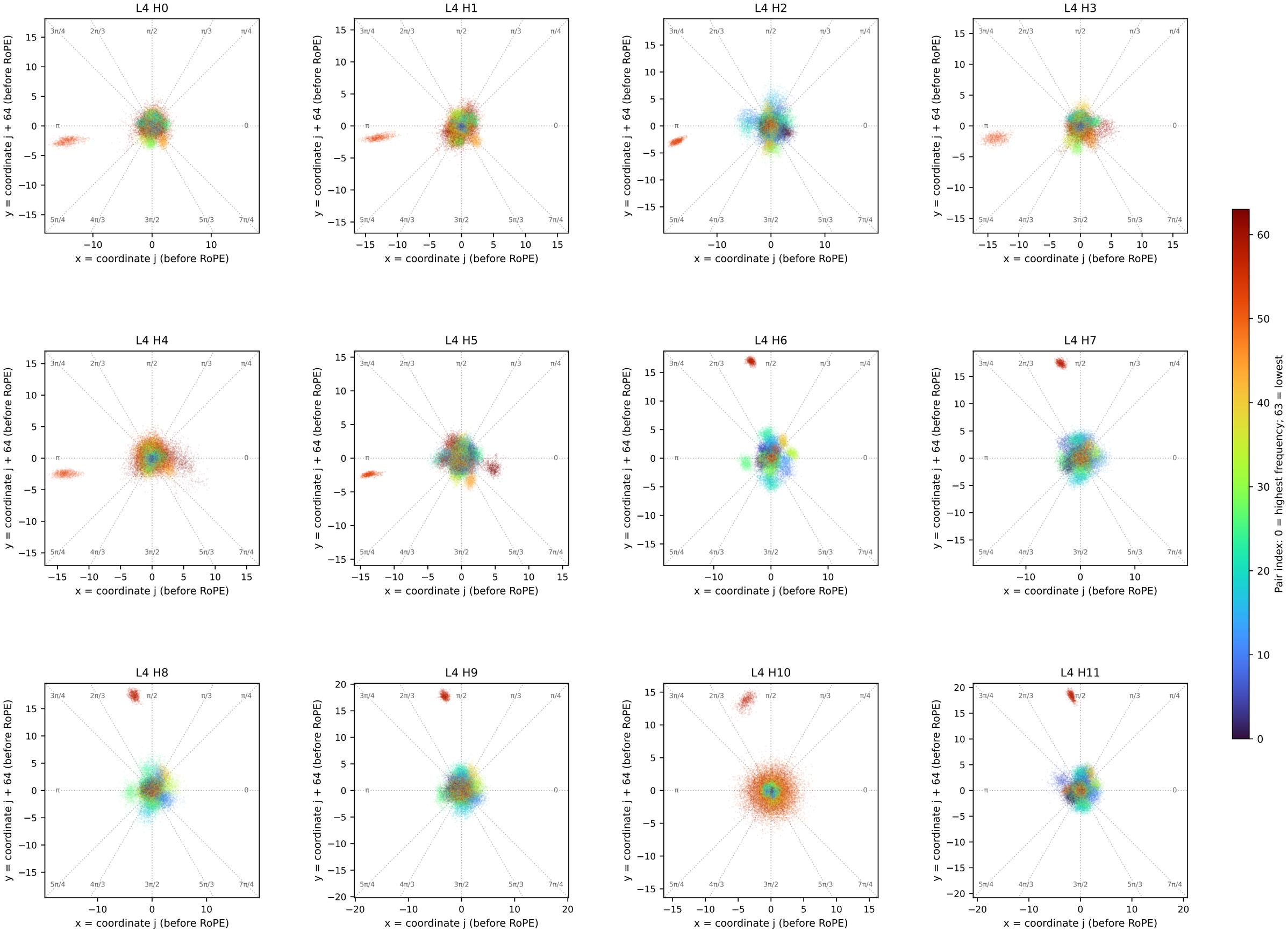
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 3 · axes fitted separately for each head



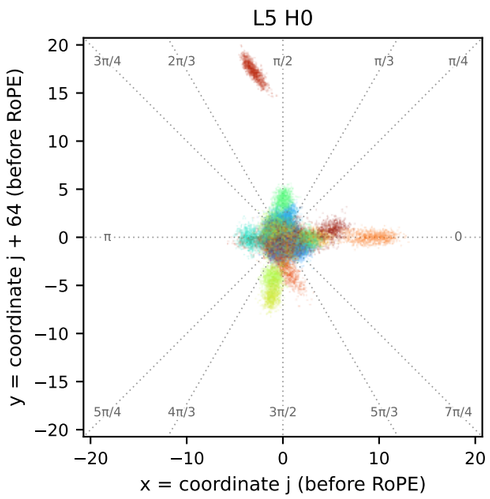
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

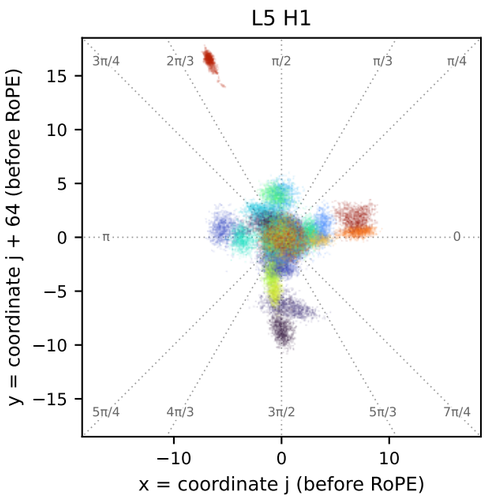
Qwen queries · layer 4 · axes fitted separately for each head

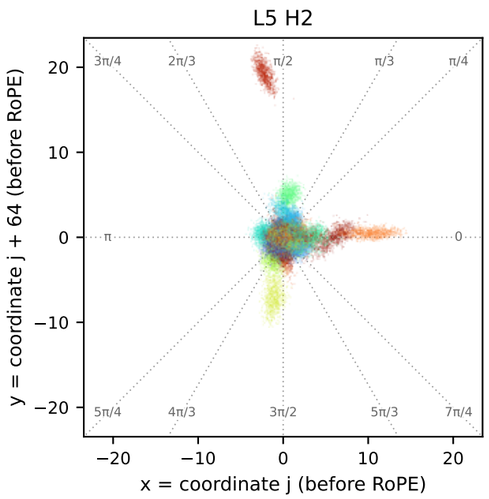


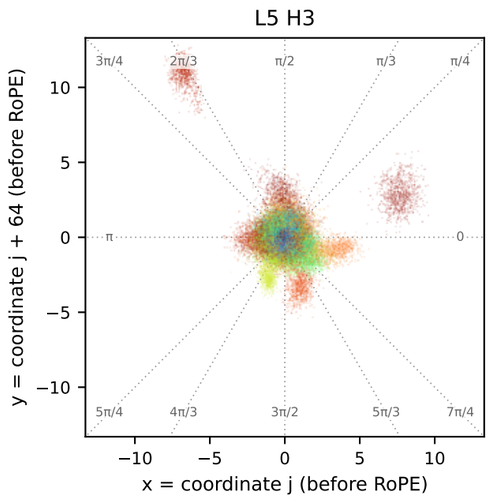
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

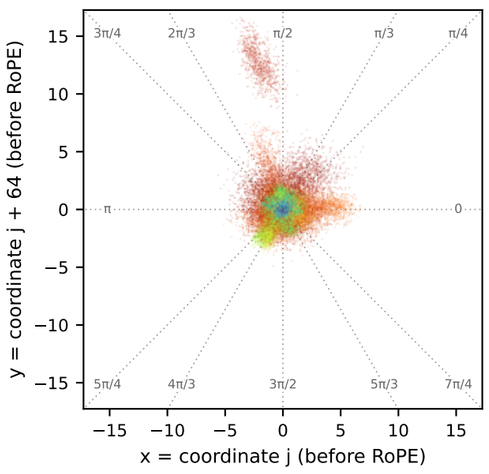
Qwen queries · layer 5 · axes fitted separately for each head

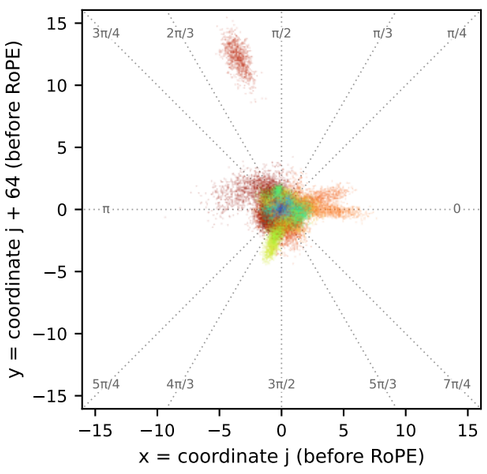


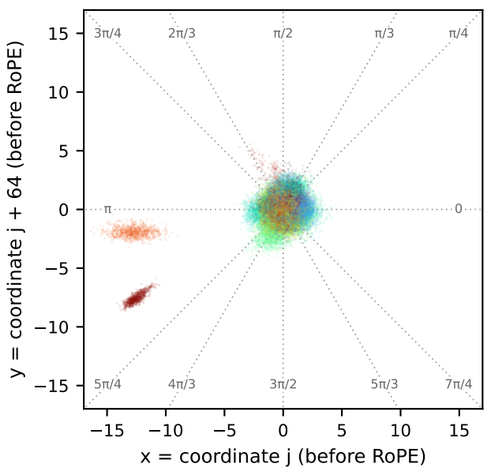


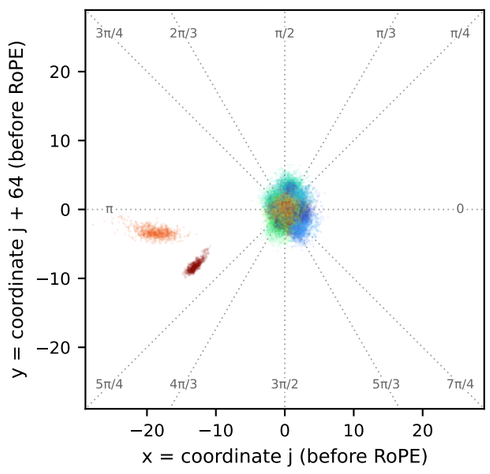


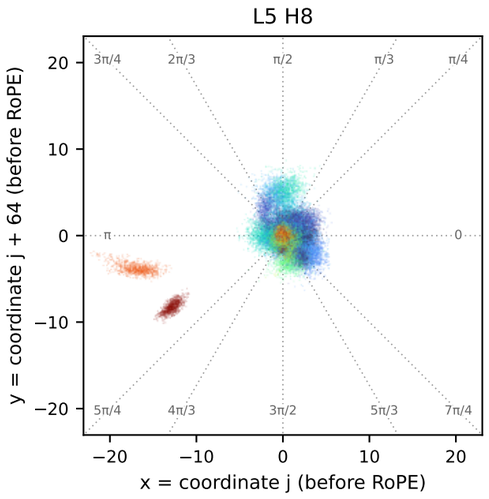


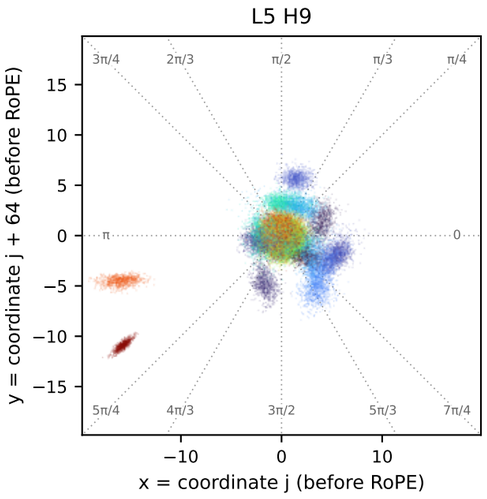


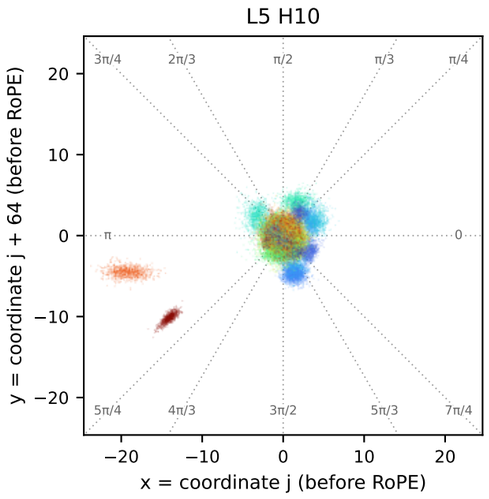


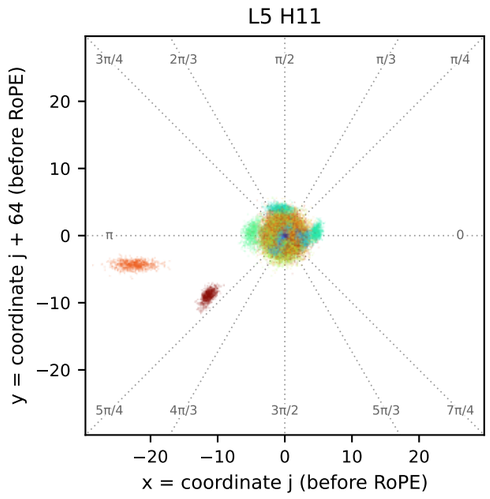


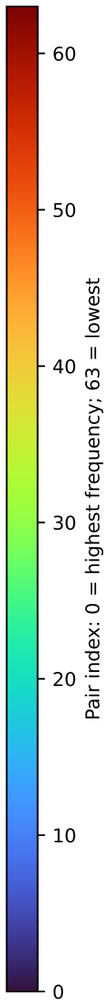






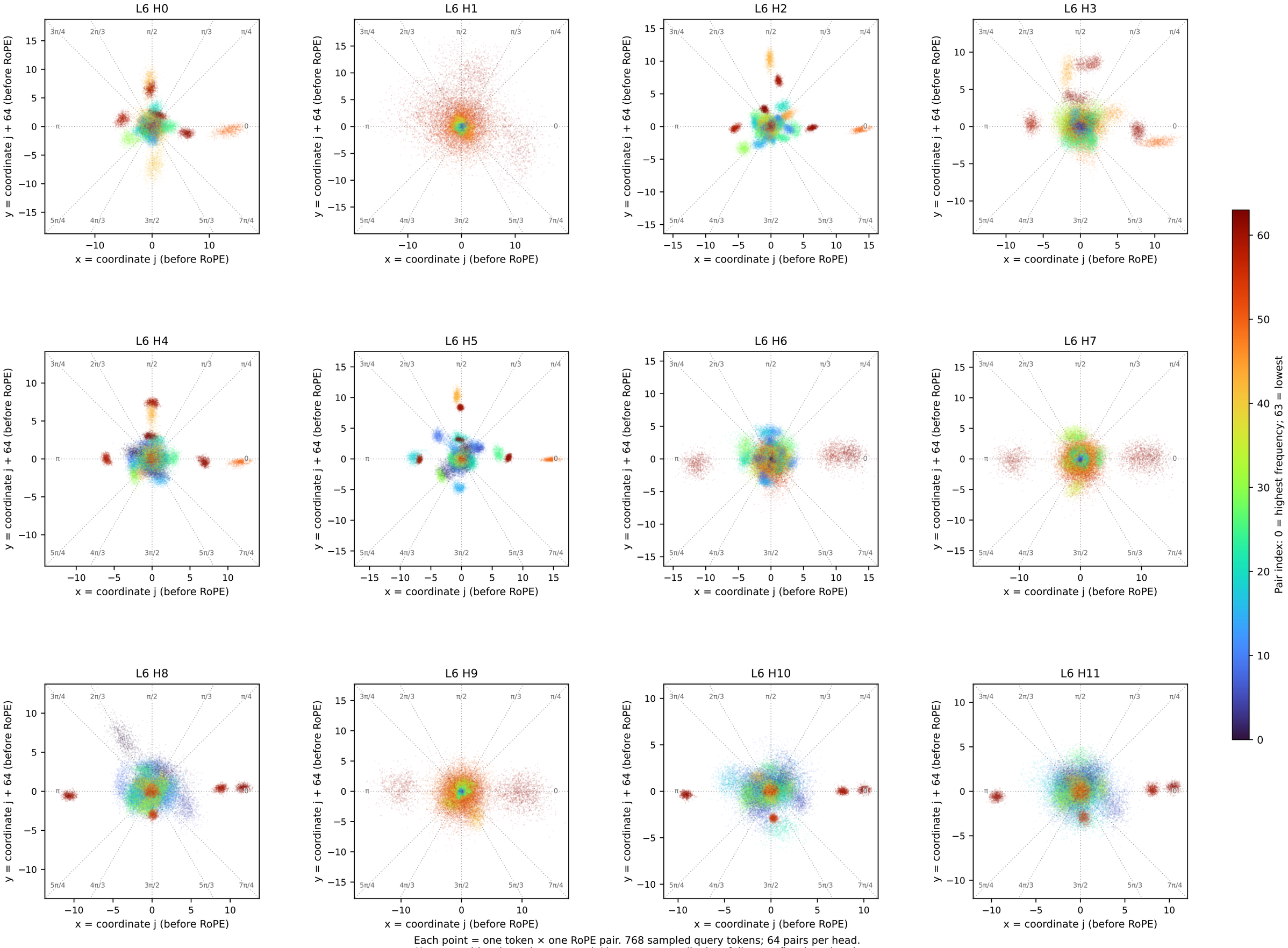






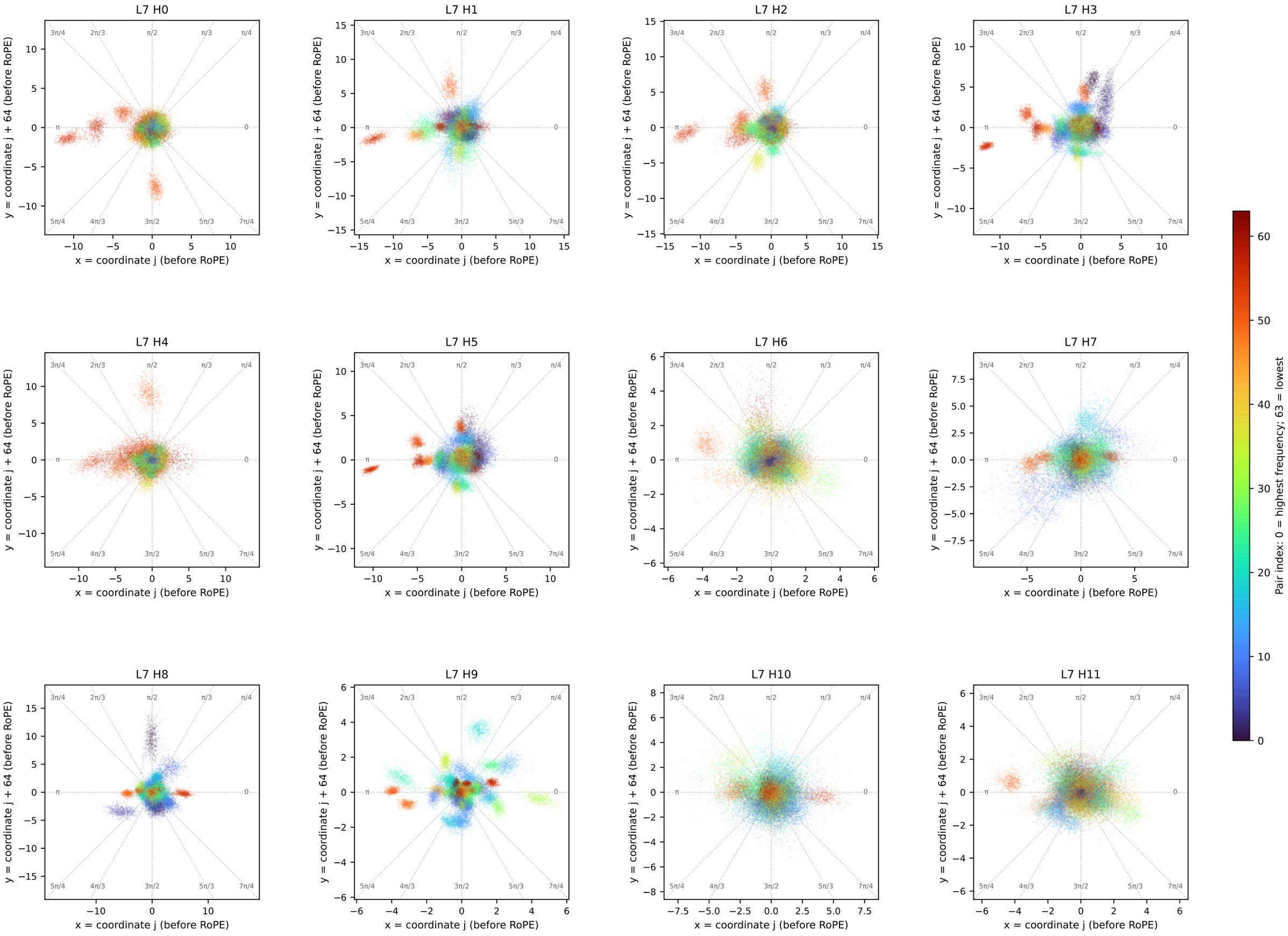
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 6 · axes fitted separately for each head



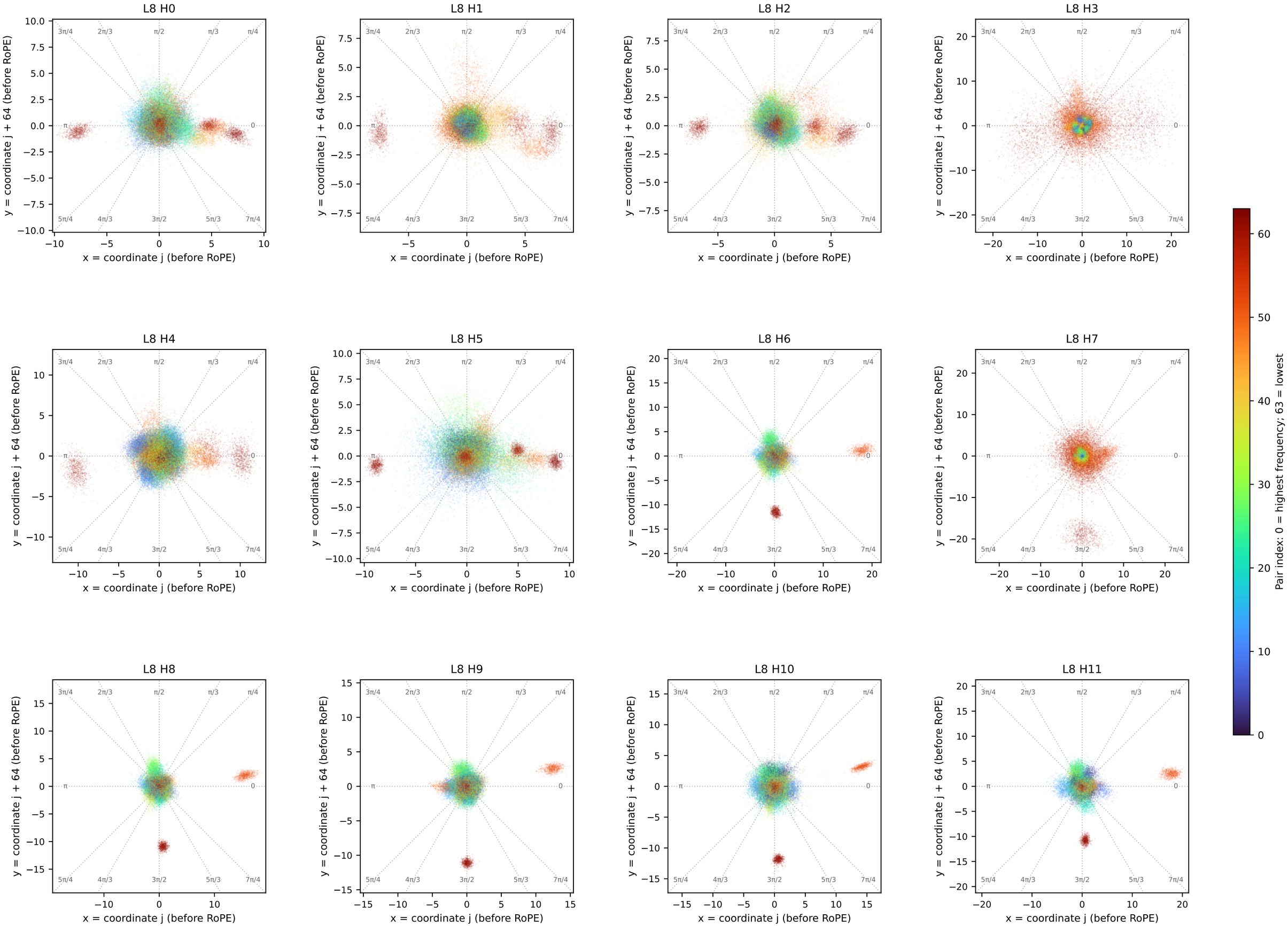
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 7 · axes fitted separately for each head



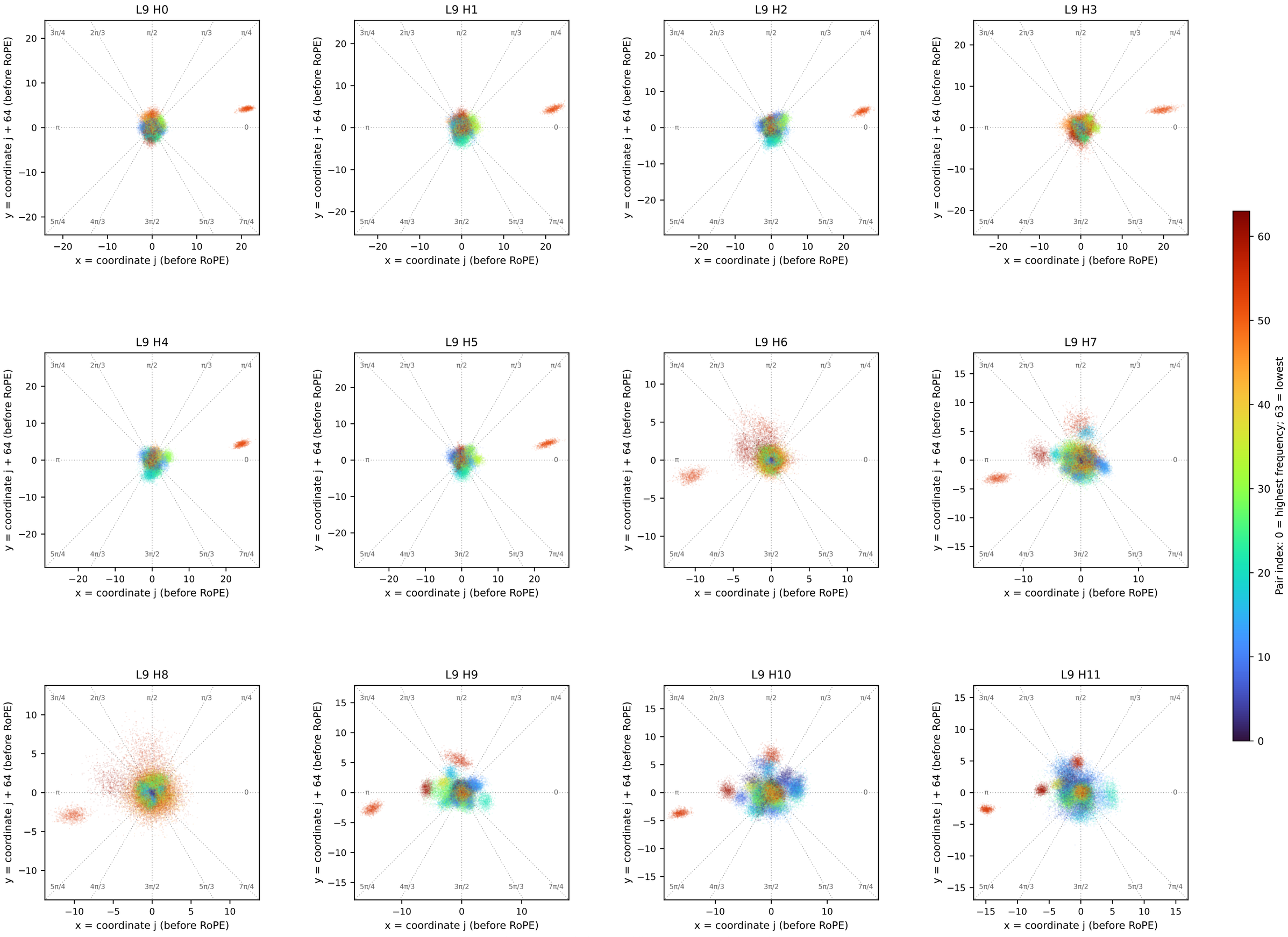
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 8 · axes fitted separately for each head



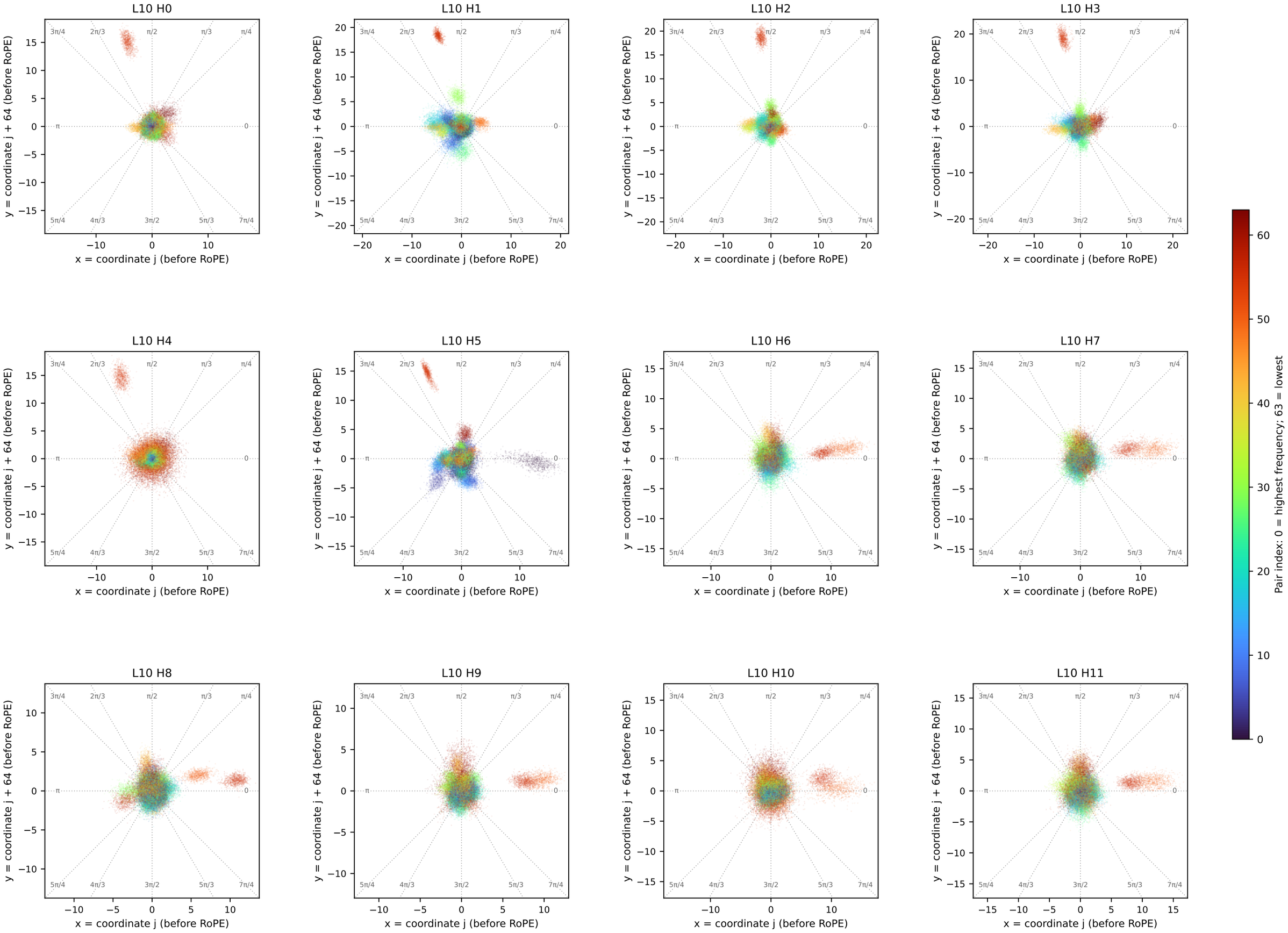
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 9 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 10 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

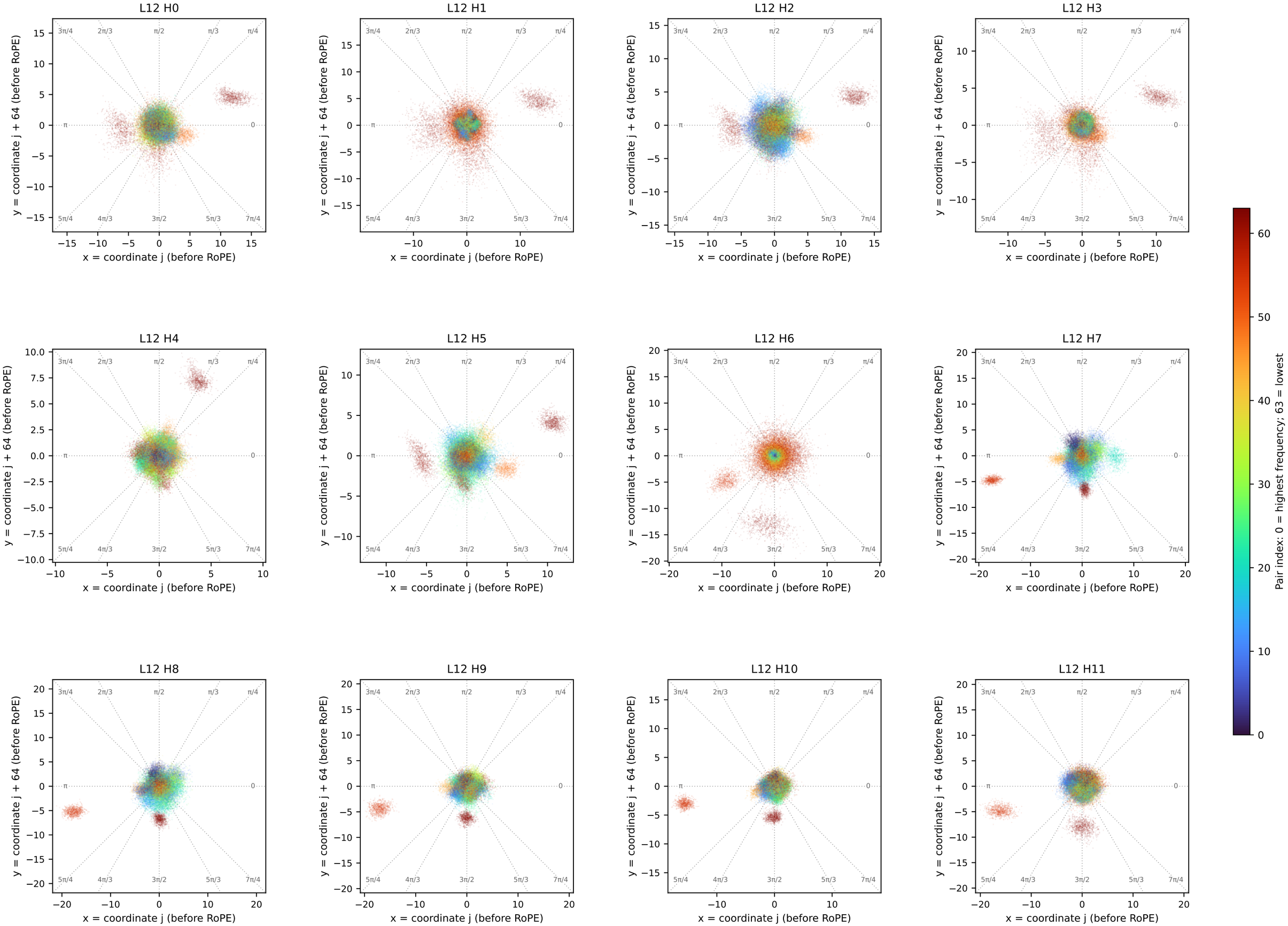
Qwen queries · layer 11 · axes fitted separately for each head

L11 H0
L11 H1
L11 H2
L11 H3
L11 H4
L11 H5
L11 H6
L11 H7
L11 H8
L11 H9
L11 H10
L11 H11

x = coordinate j (before RoPE)
y = coordinate j + 64 (before RoPE)
Pair index: 0 = highest frequency; 63 = lowest

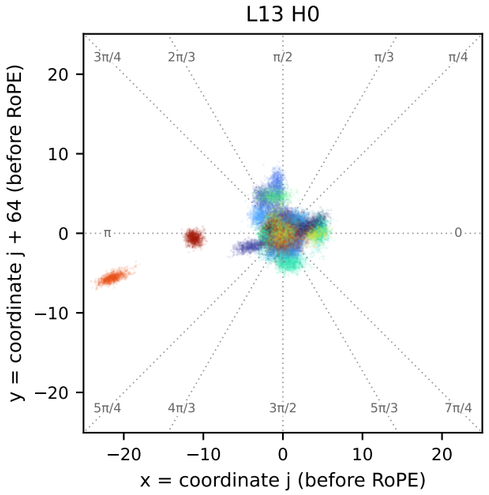
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

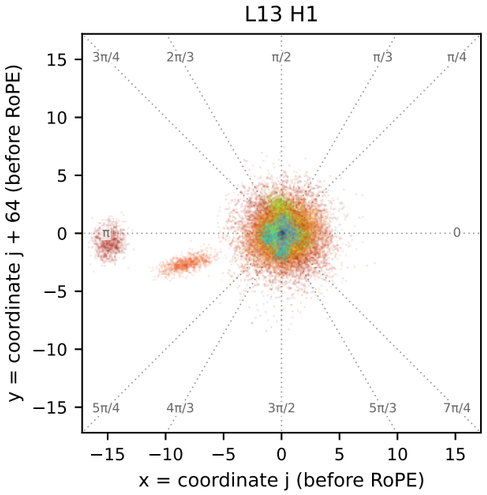
Qwen queries · layer 12 · axes fitted separately for each head

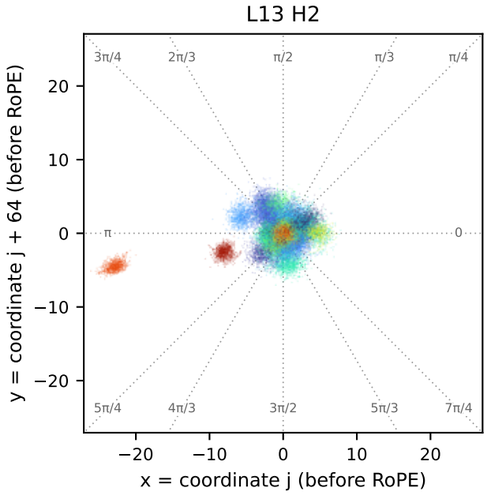


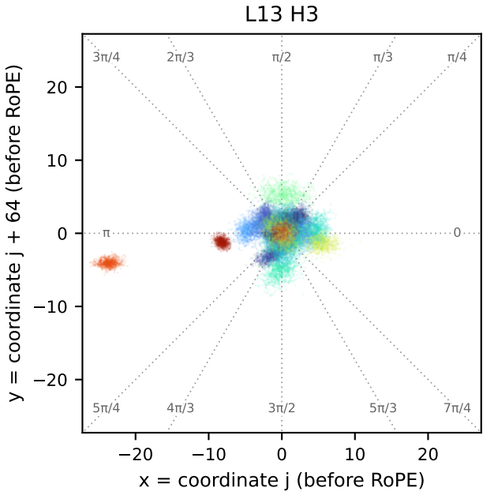
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

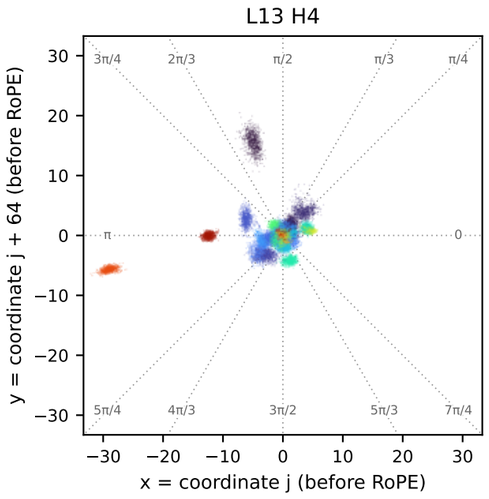
Qwen queries · layer 13 · axes fitted separately for each head

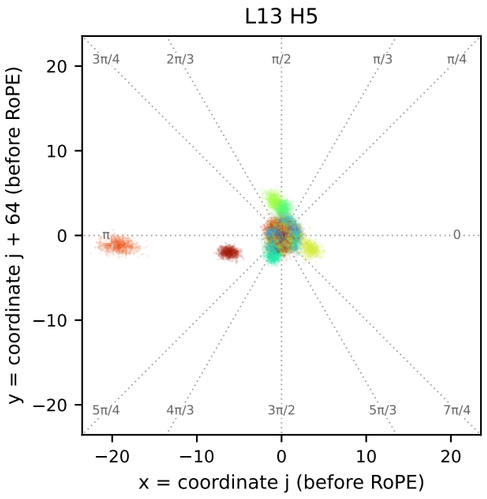


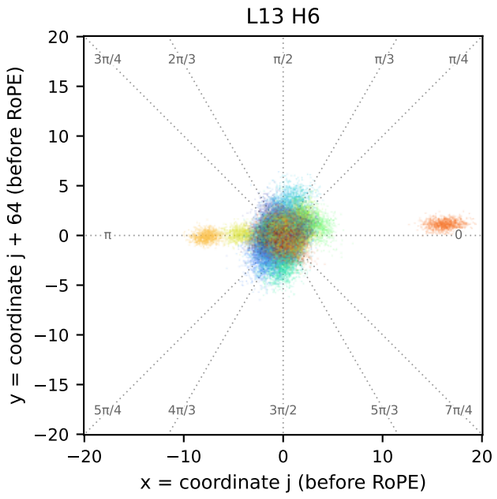


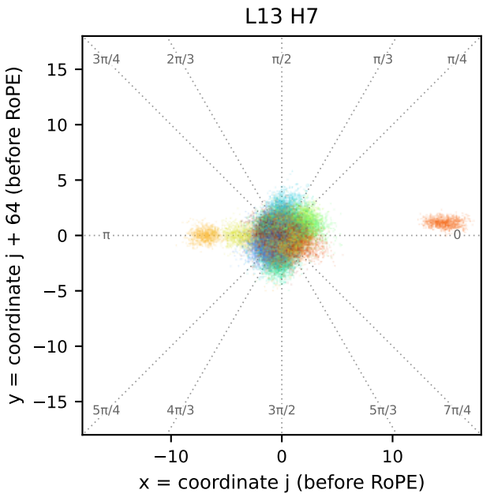


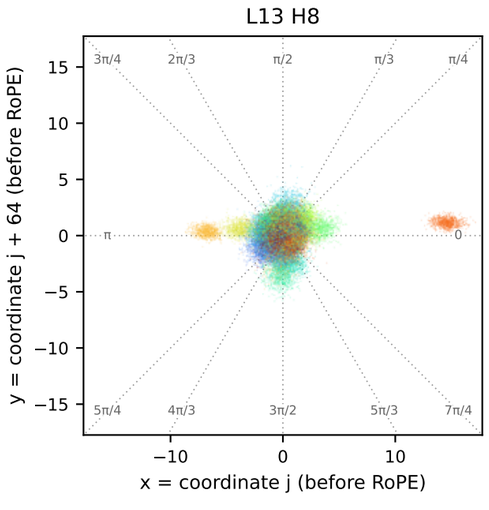


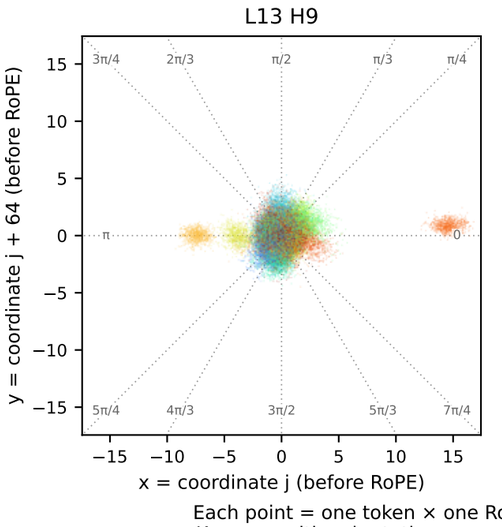


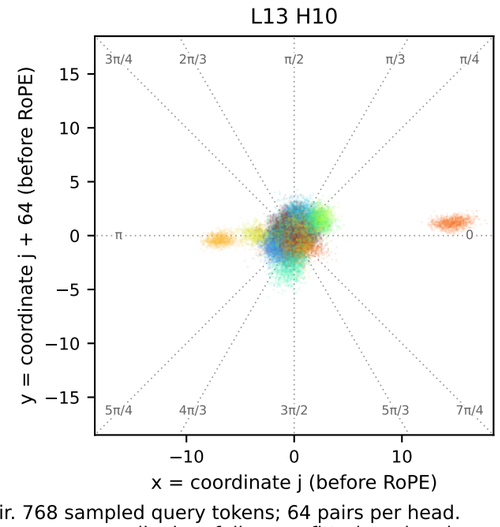


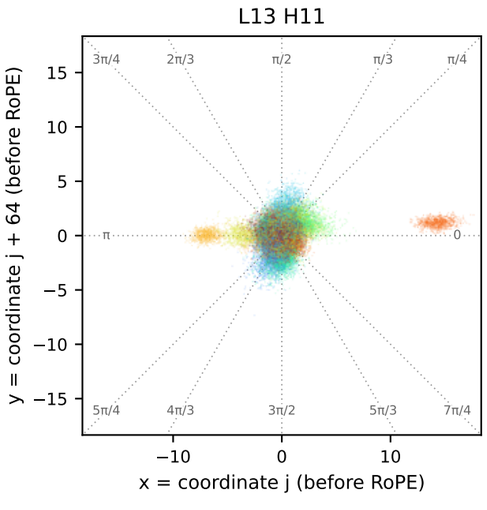


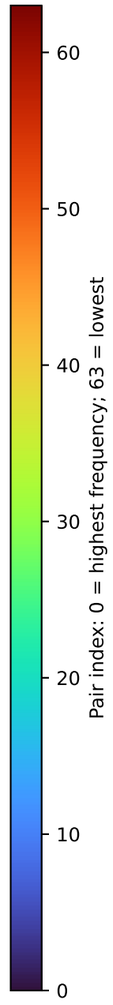






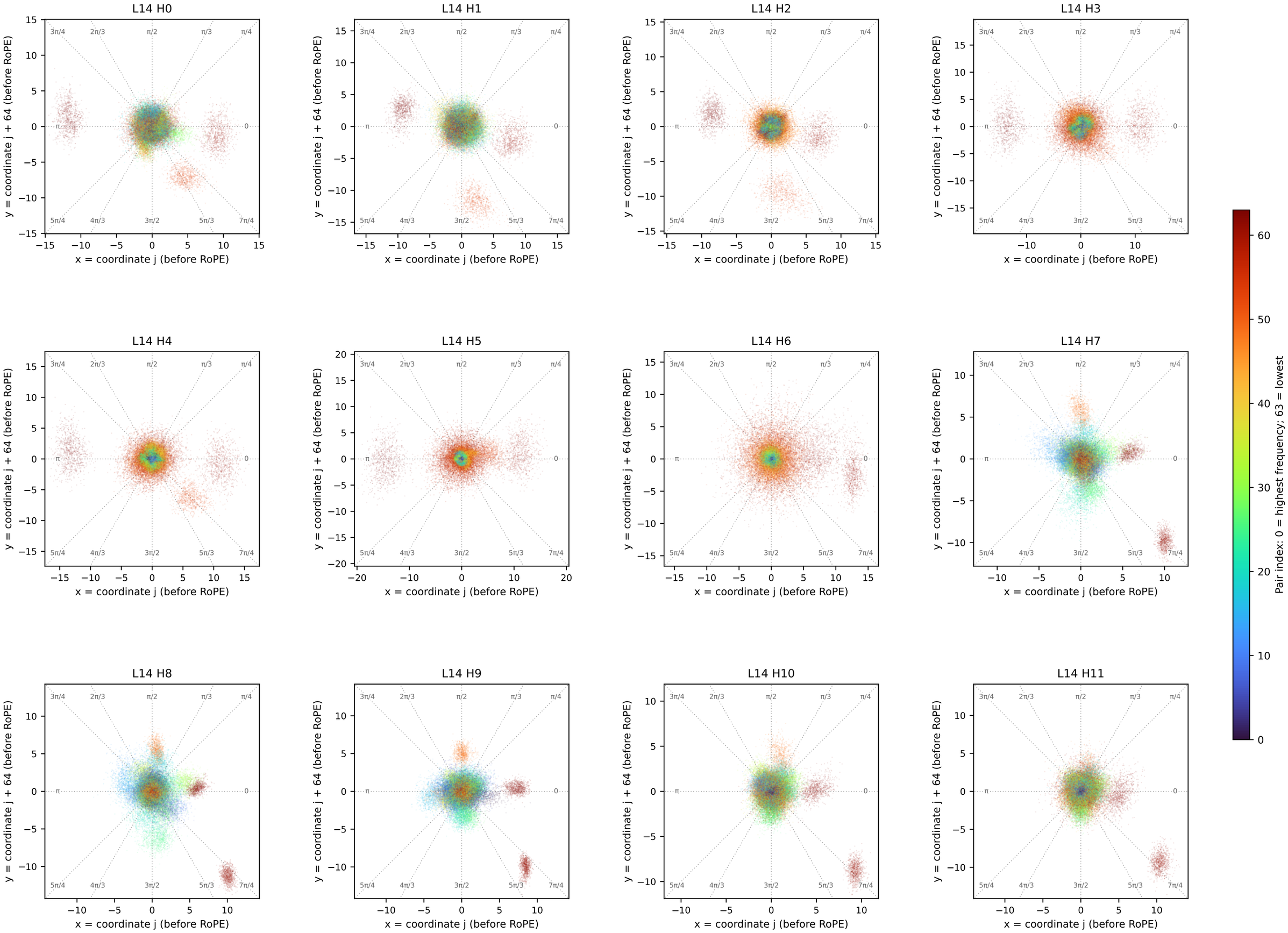






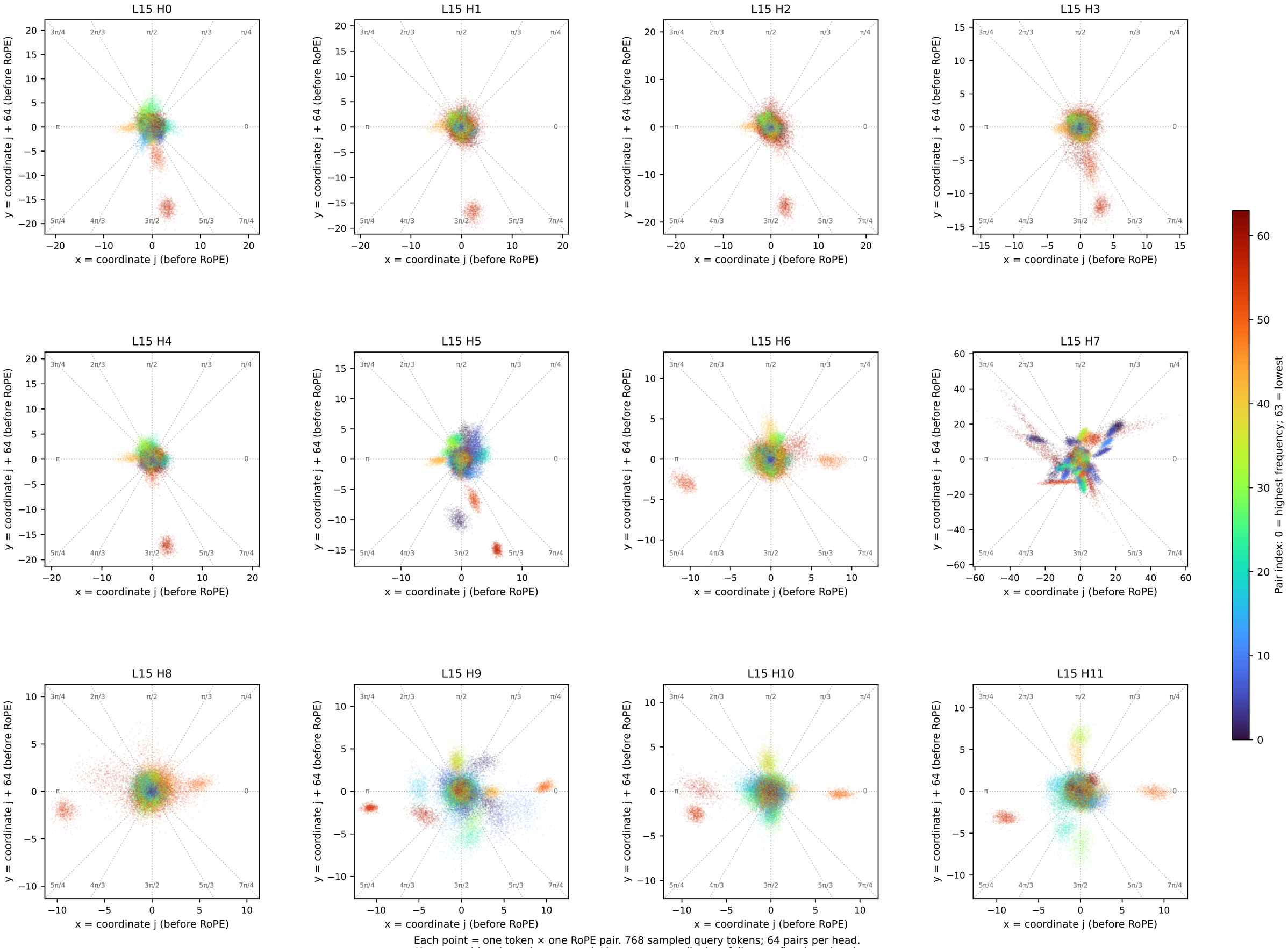
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 14 · axes fitted separately for each head



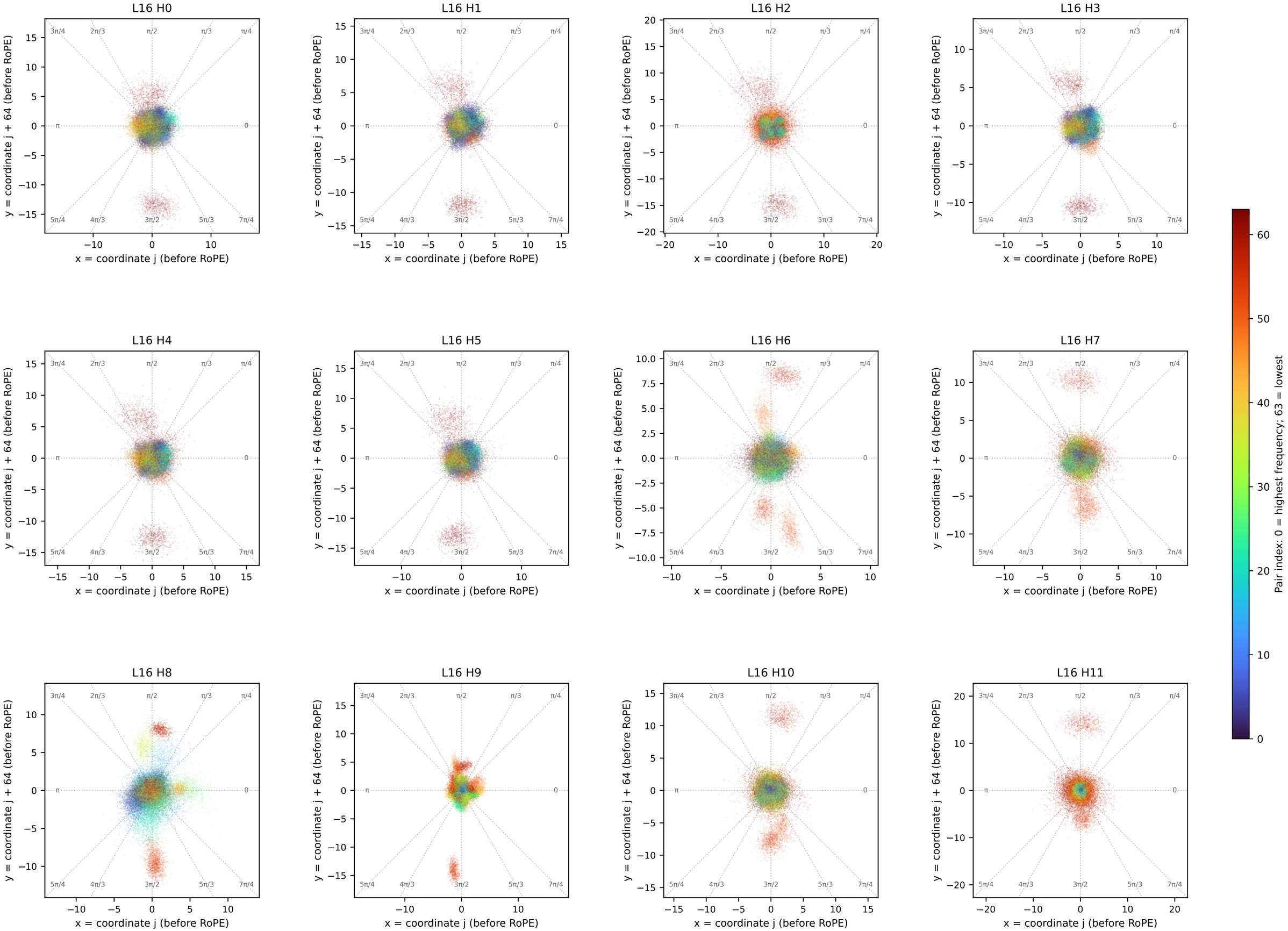
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 15 · axes fitted separately for each head



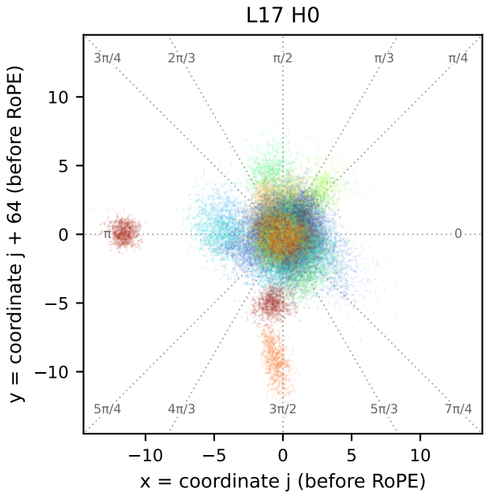
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

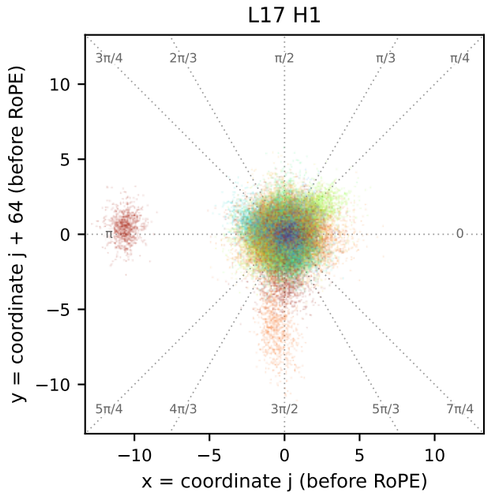
Qwen queries · layer 16 · axes fitted separately for each head

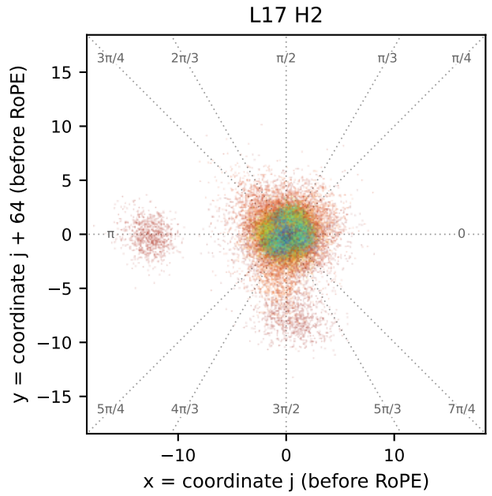


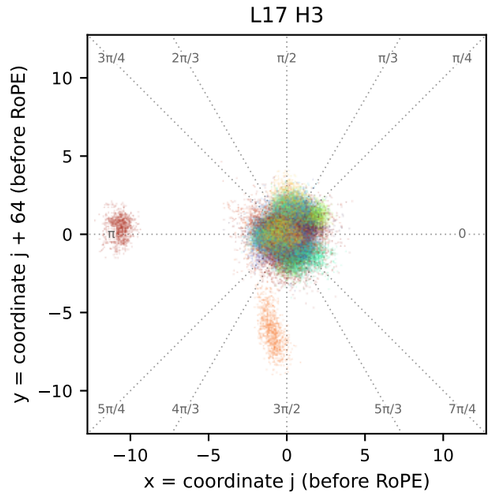
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

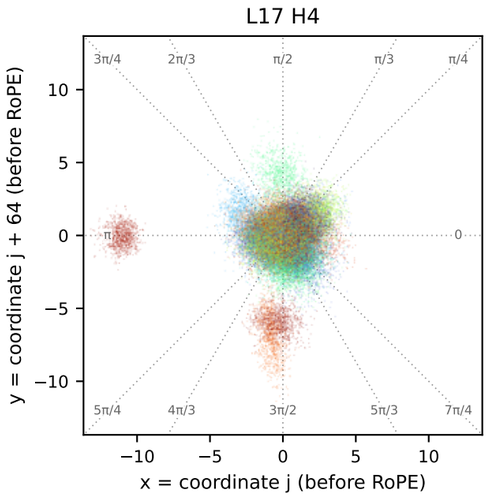
Qwen queries · layer 17 · axes fitted separately for each head

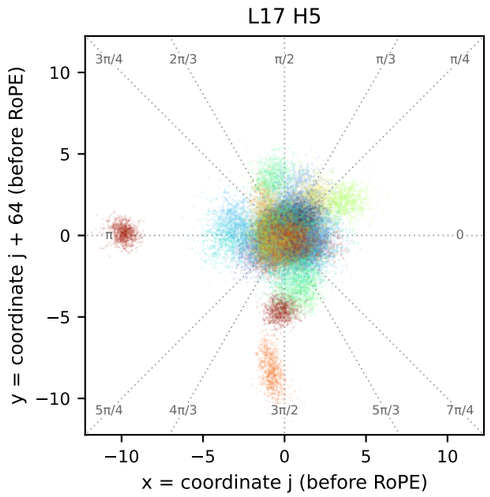


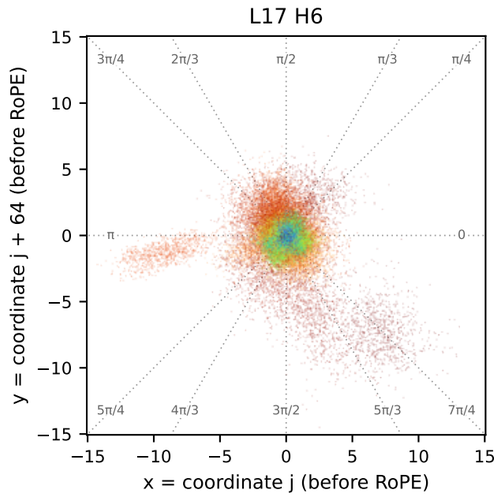


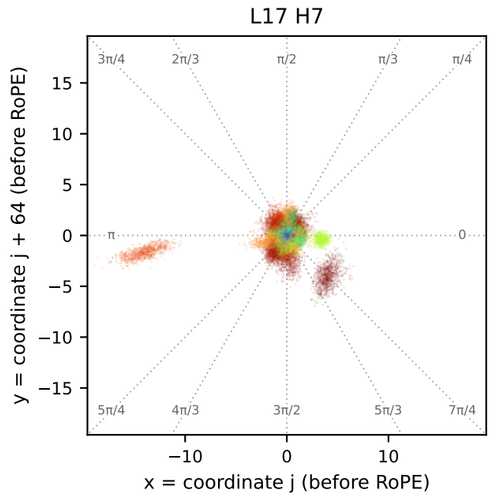


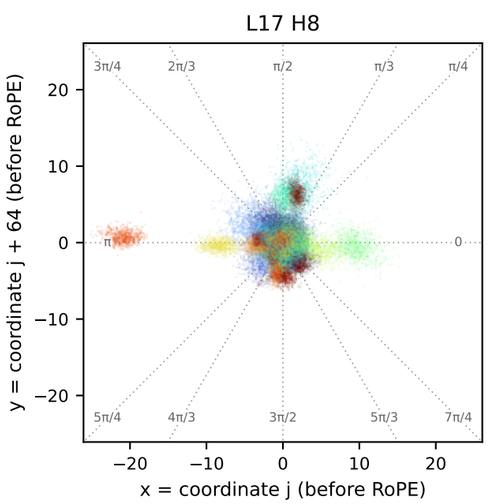


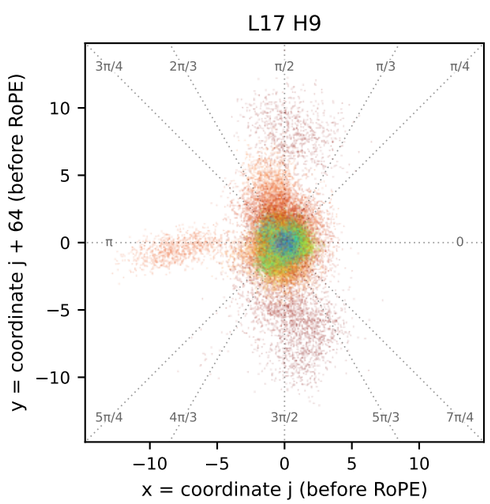


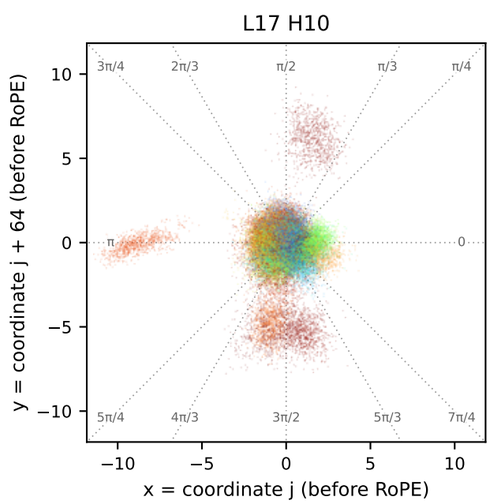


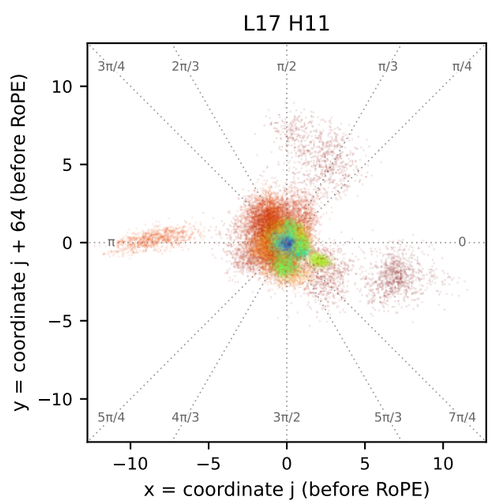


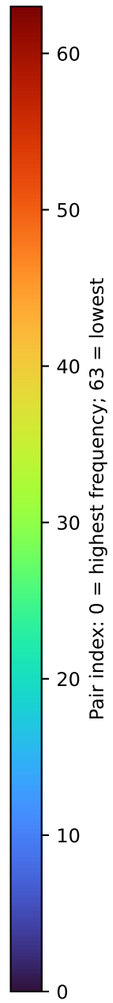






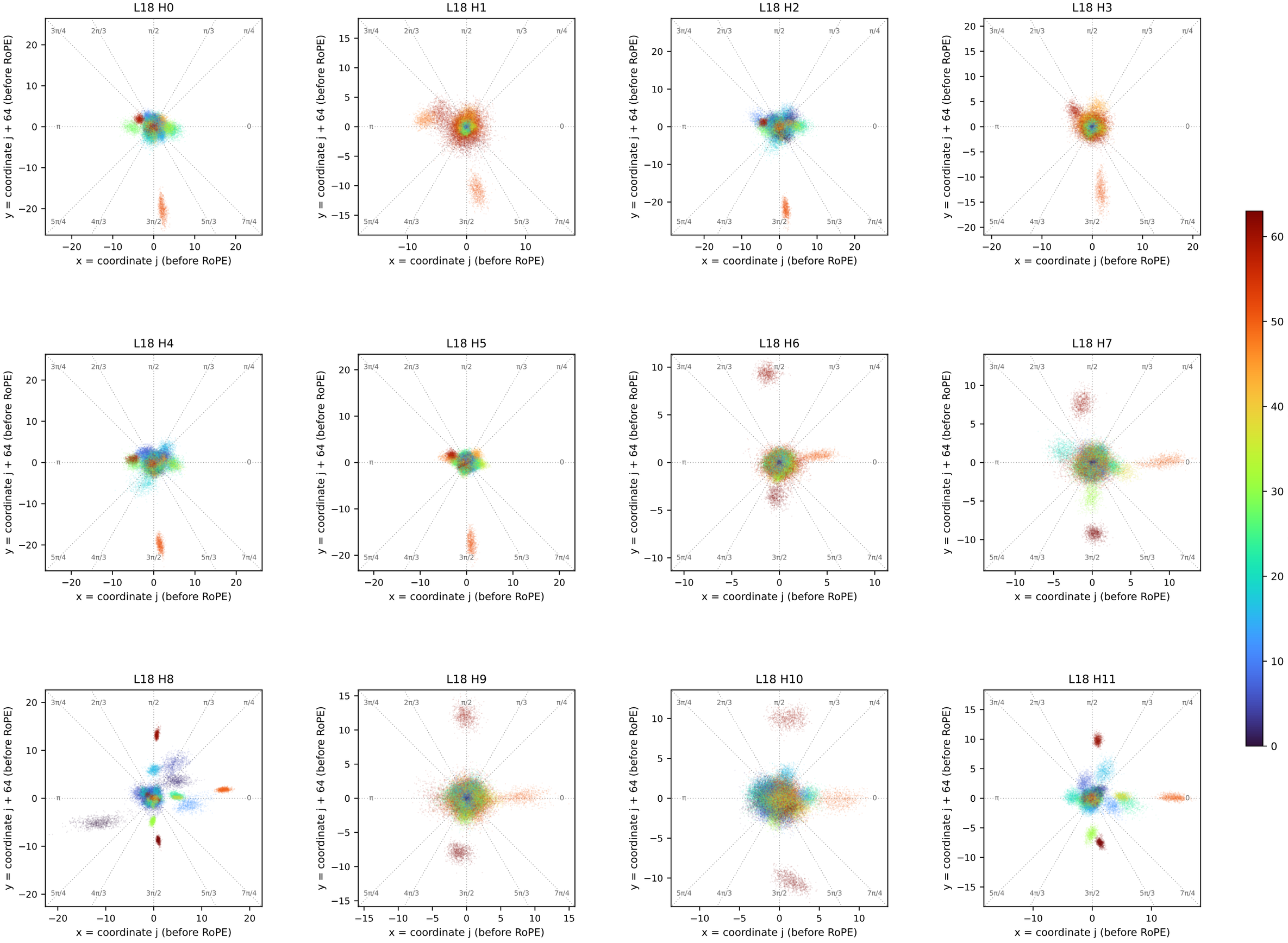






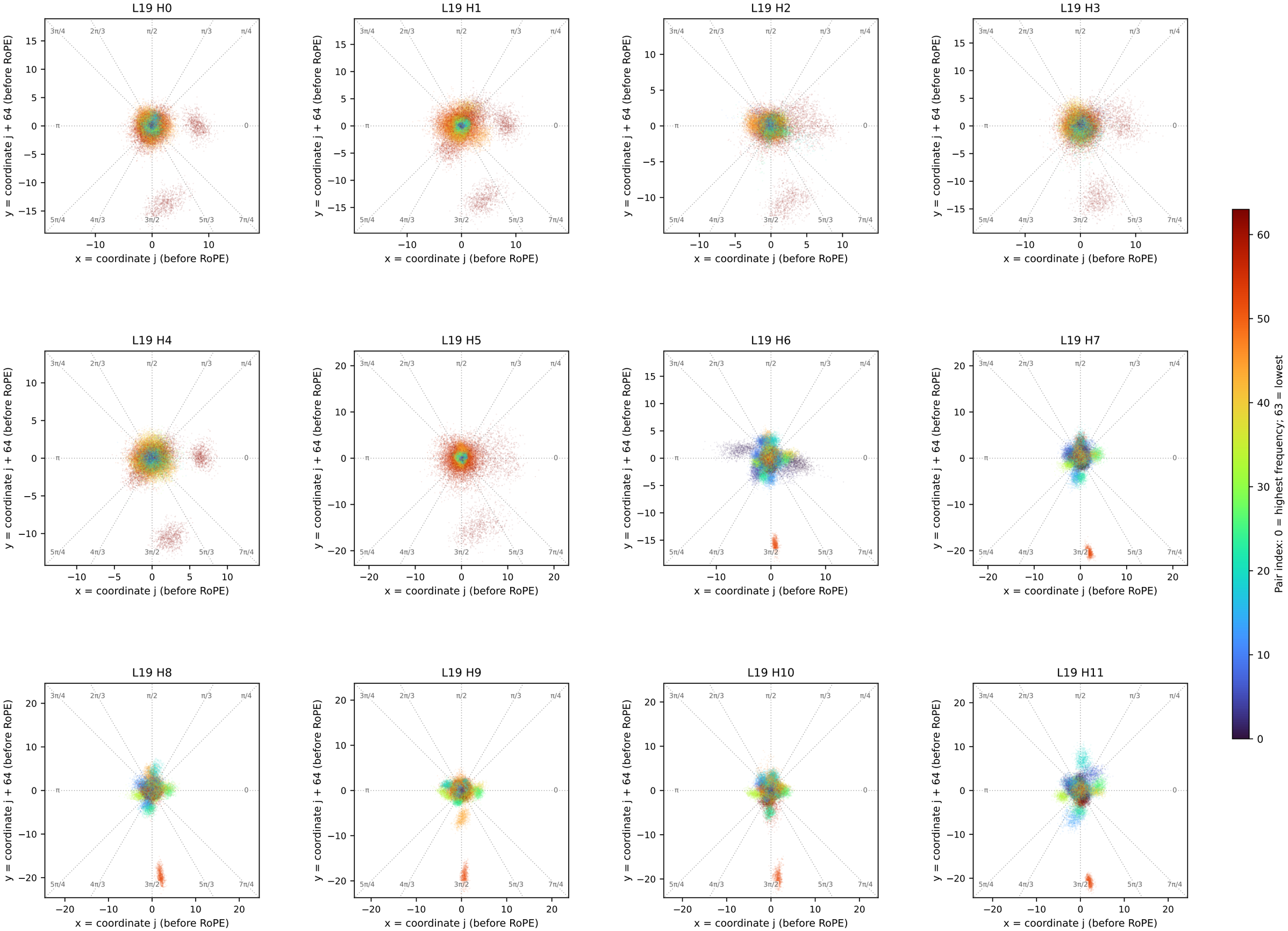
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 18 · axes fitted separately for each head



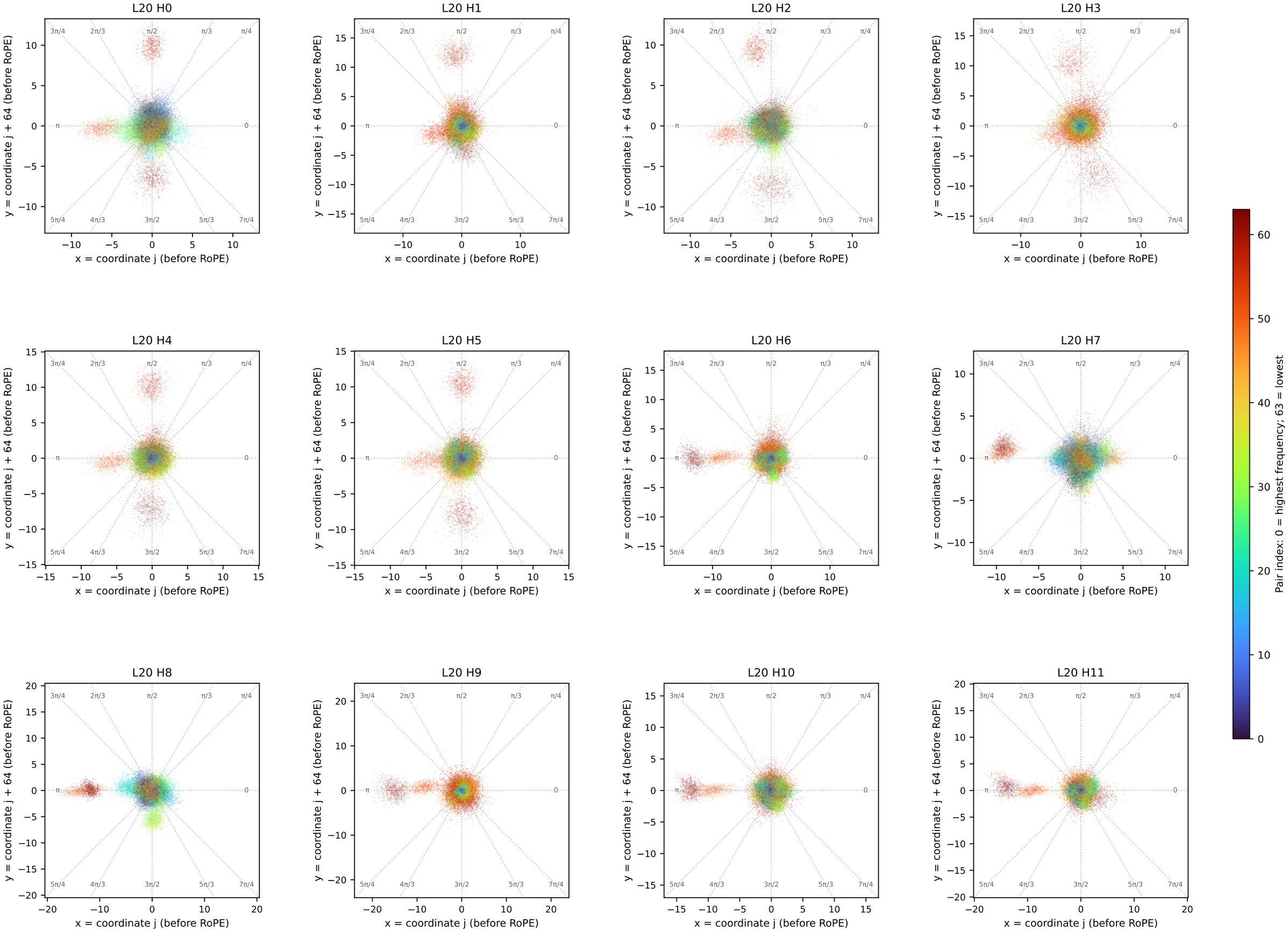
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 19 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

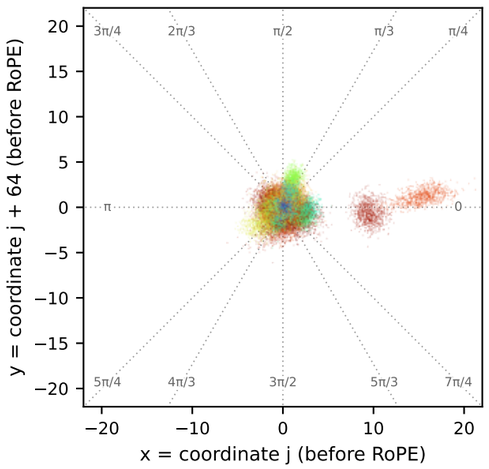
Qwen queries · layer 20 · axes fitted separately for each head



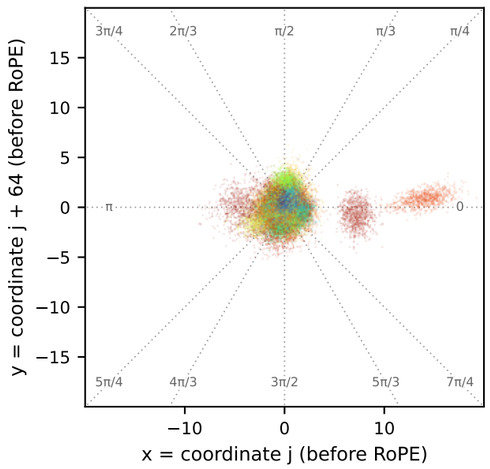
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 21 · axes fitted separately for each head

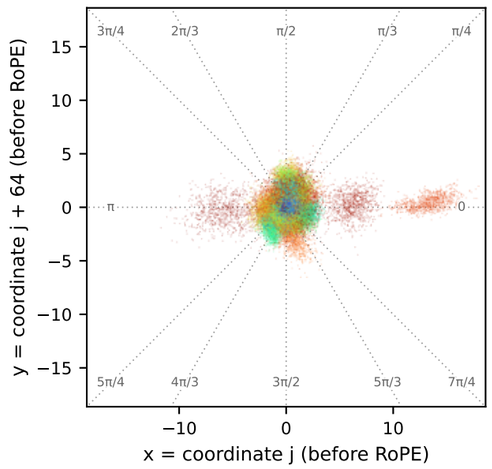
L21 H0



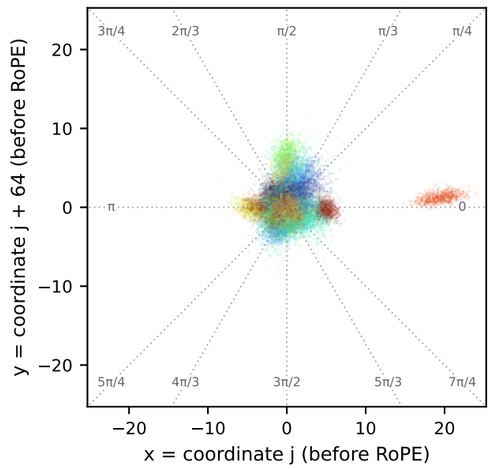
L21 H1



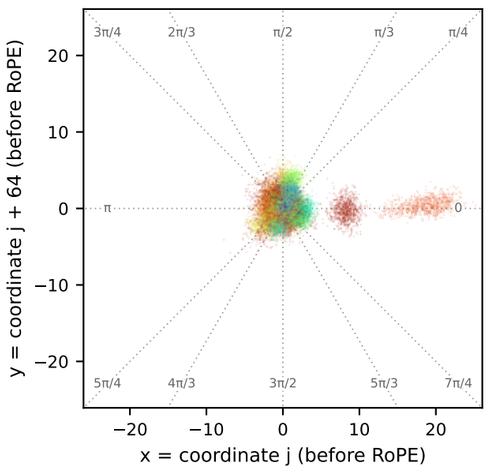
L21 H2



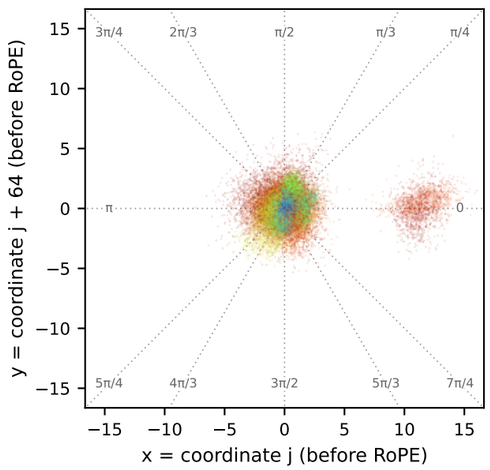
L21 H3



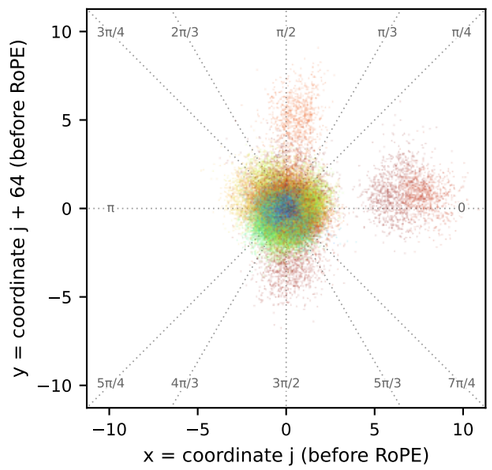
L21 H4



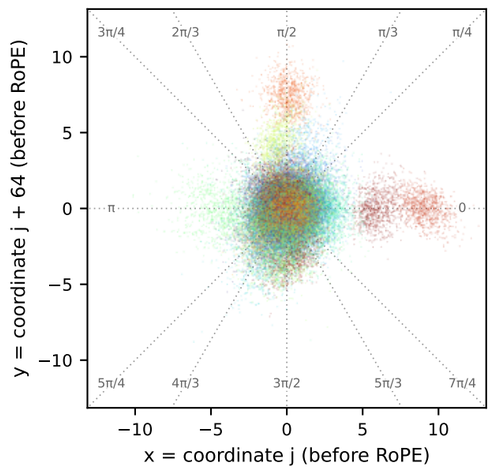
L21 H5



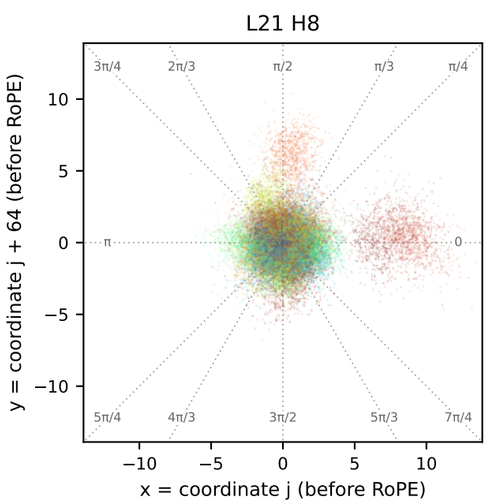
L21 H6



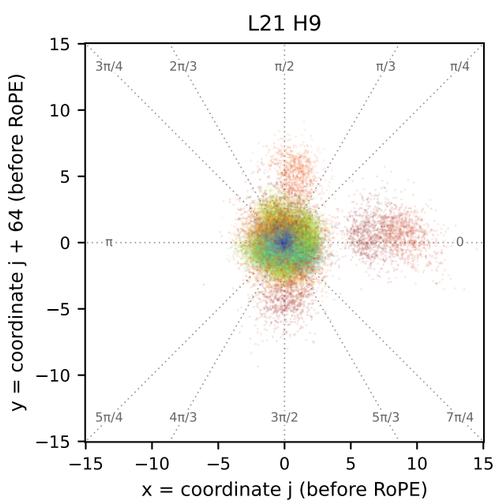
L21 H7



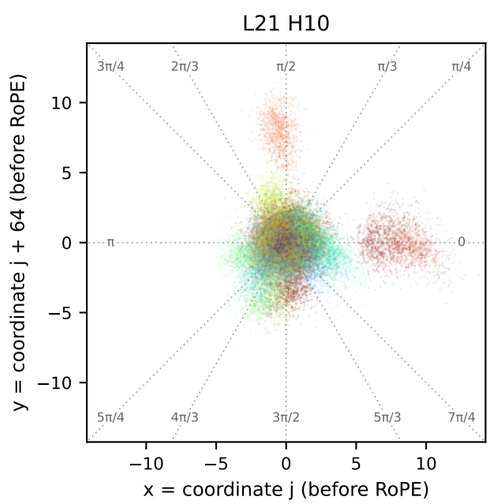
L21 H8



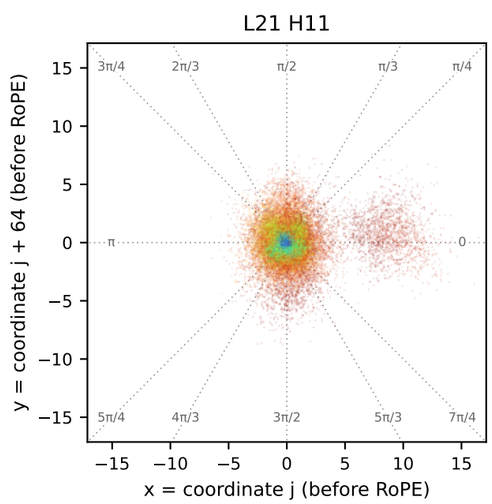
L21 H9

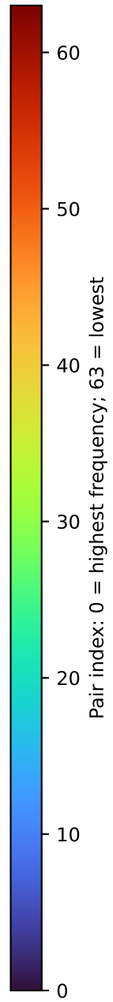


L21 H10



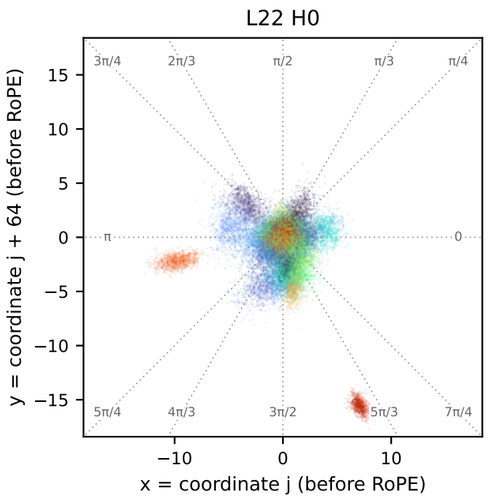
L21 H11

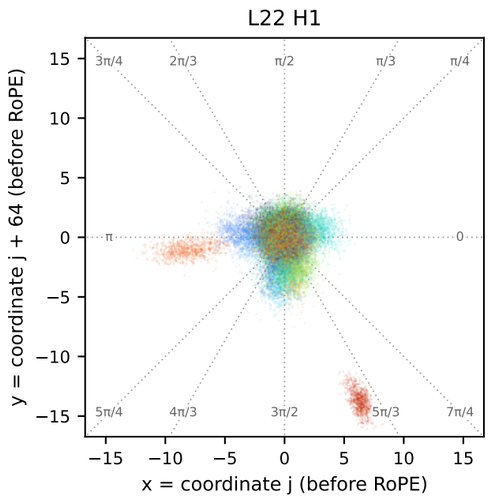


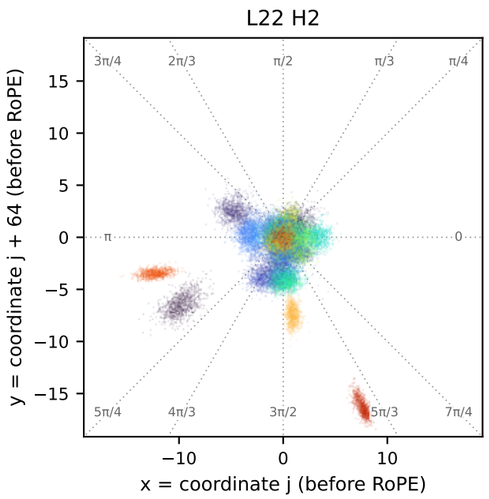


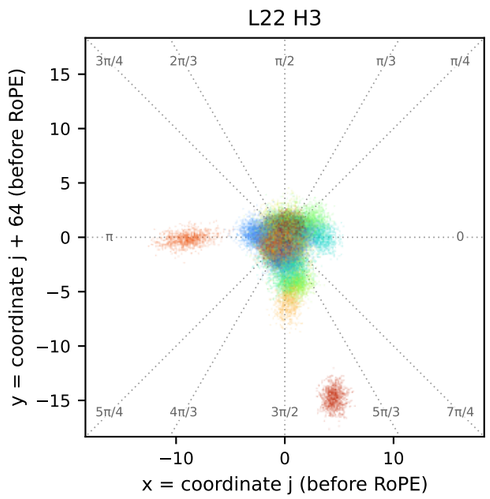
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

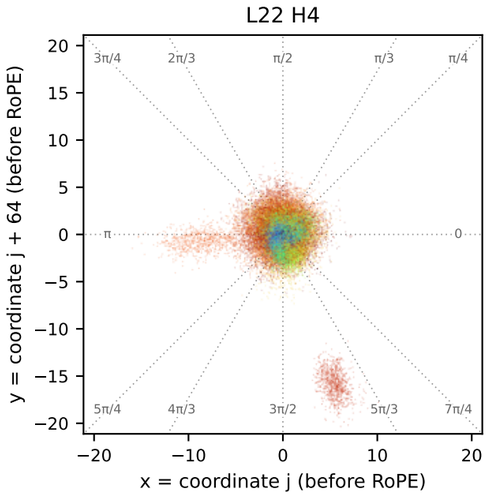
Qwen queries · layer 22 · axes fitted separately for each head

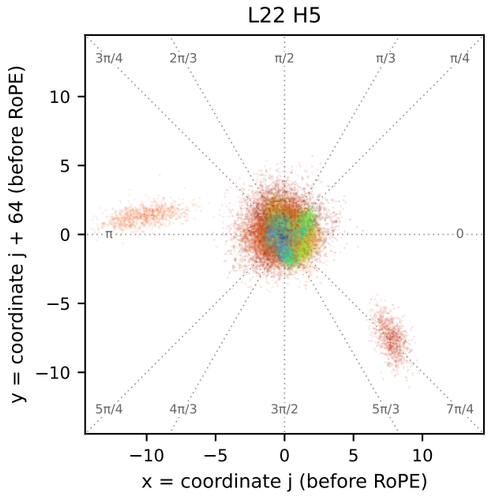


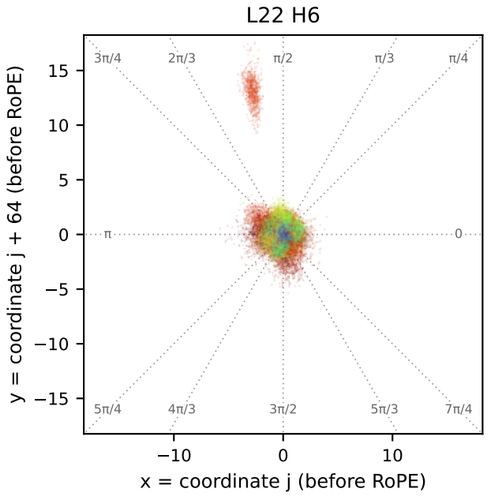


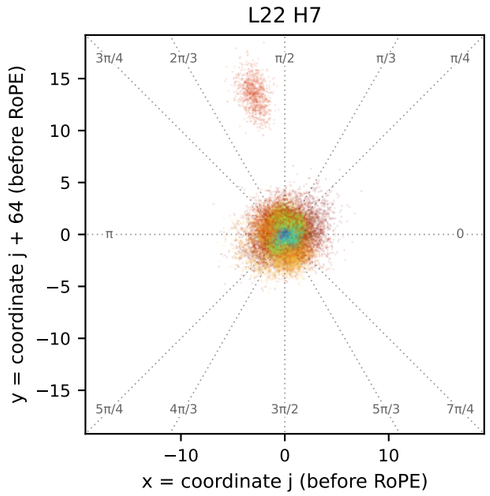


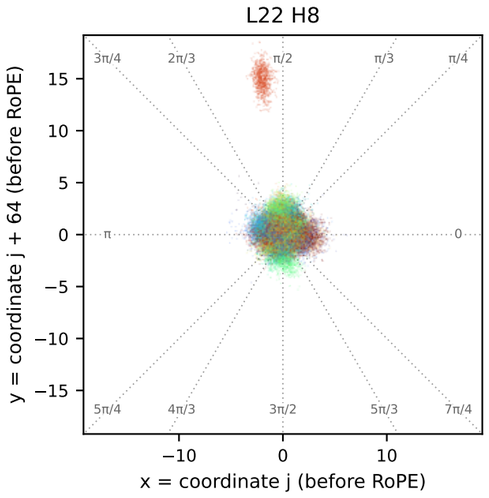


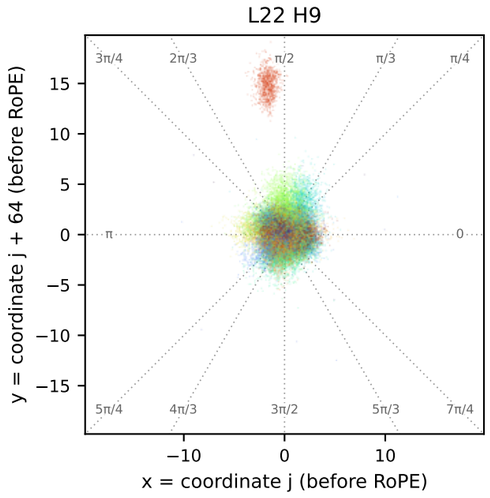


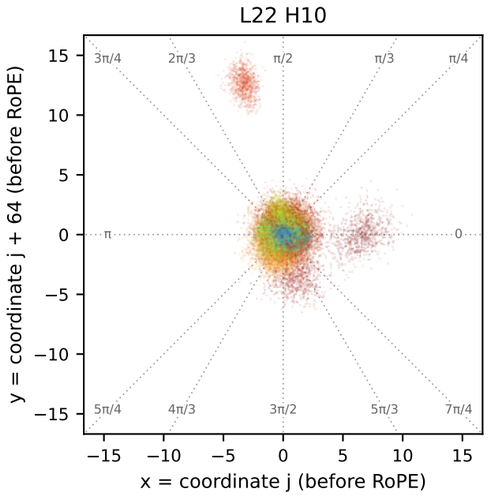


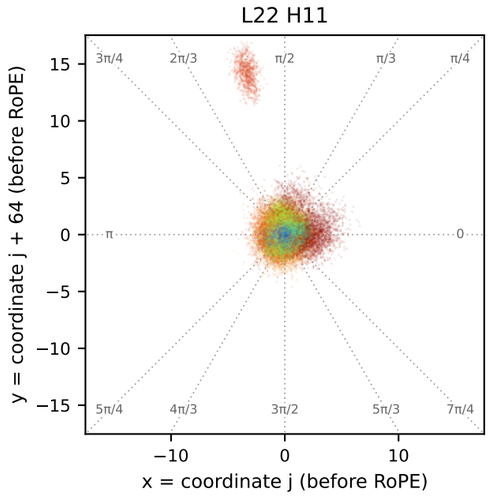


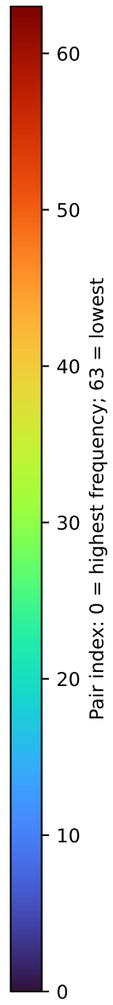






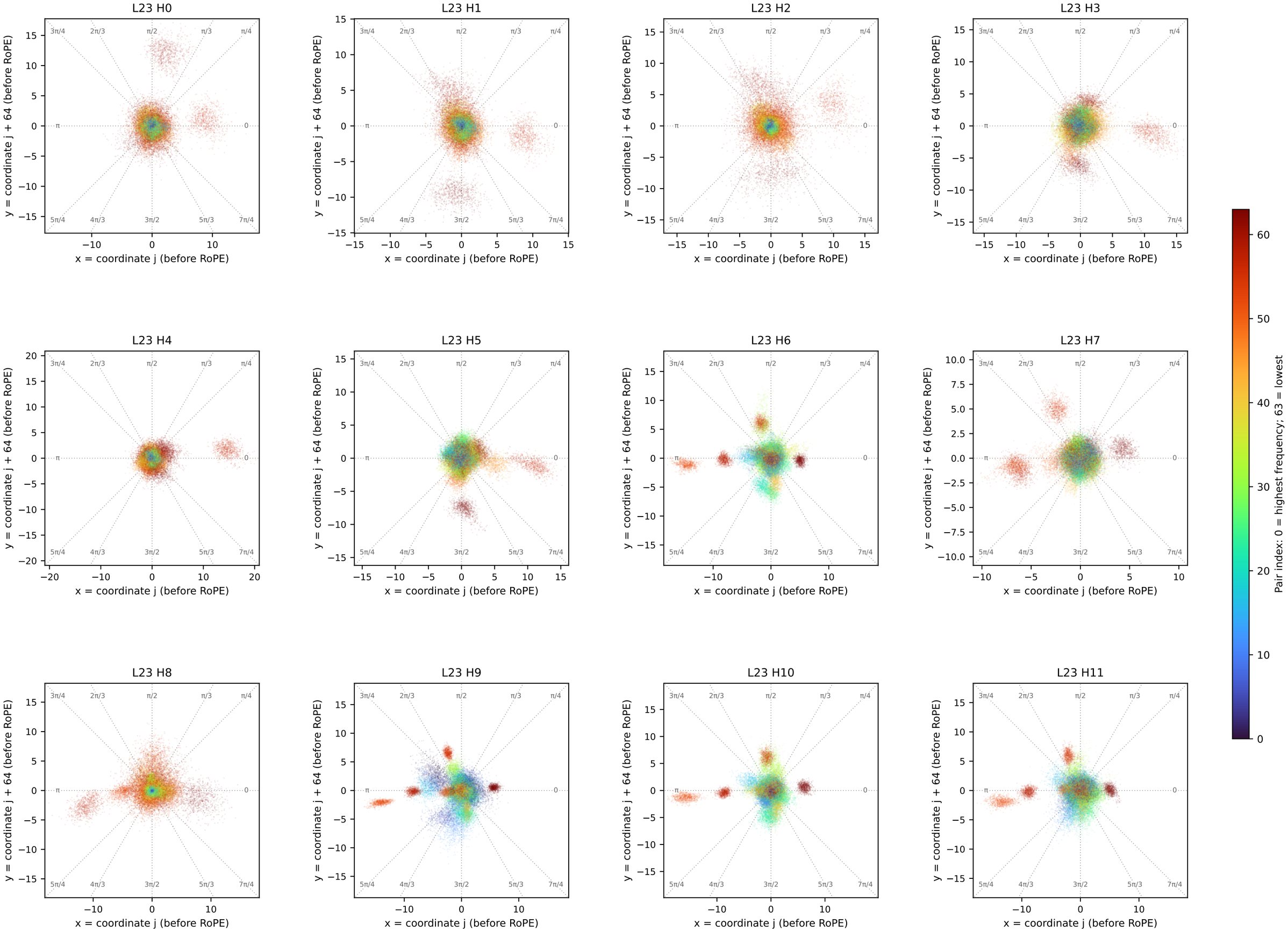






Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

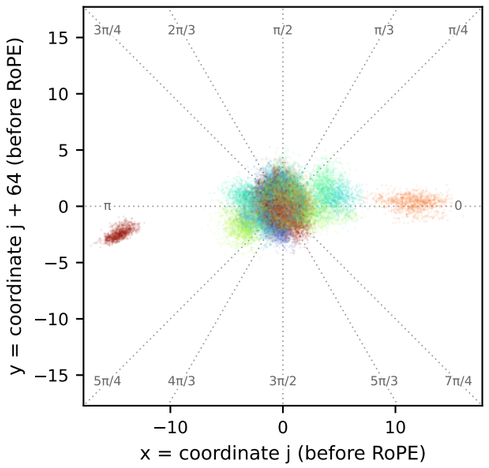
Qwen queries · layer 23 · axes fitted separately for each head



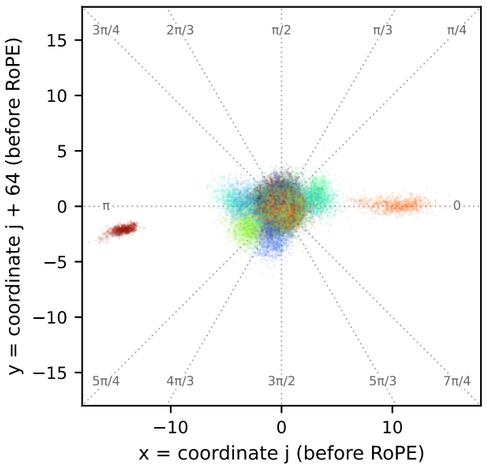
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 24 · axes fitted separately for each head

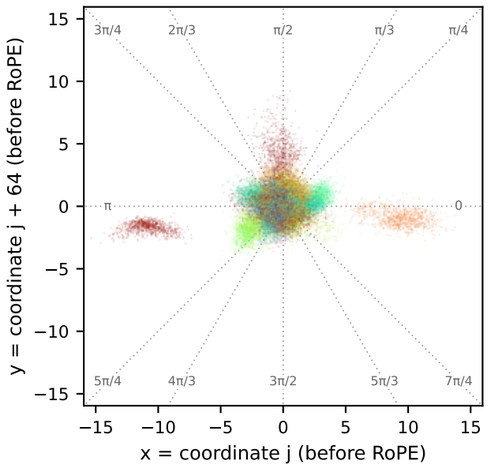
L24 H0



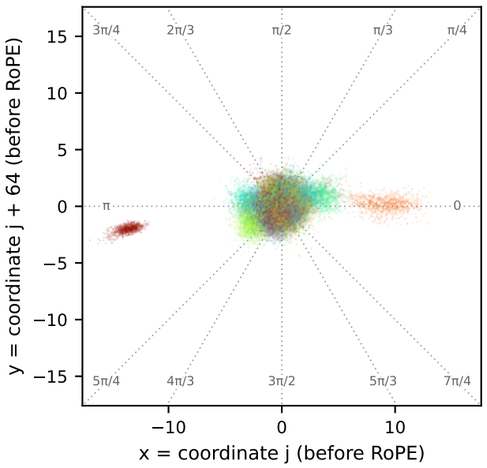
L24 H1



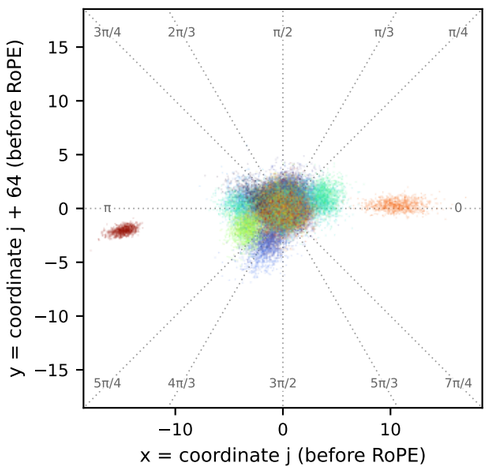
L24 H2



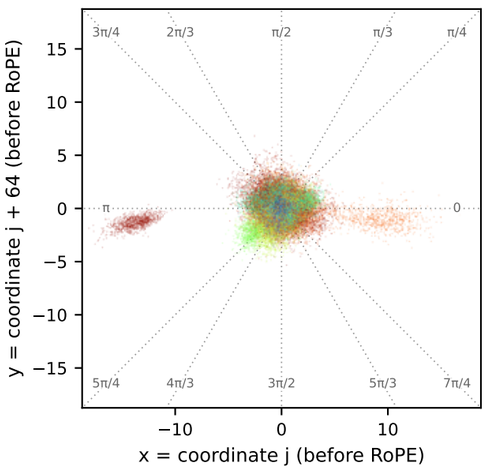
L24 H3



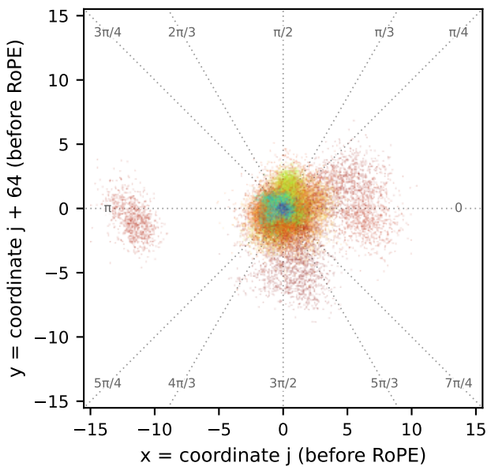
L24 H4



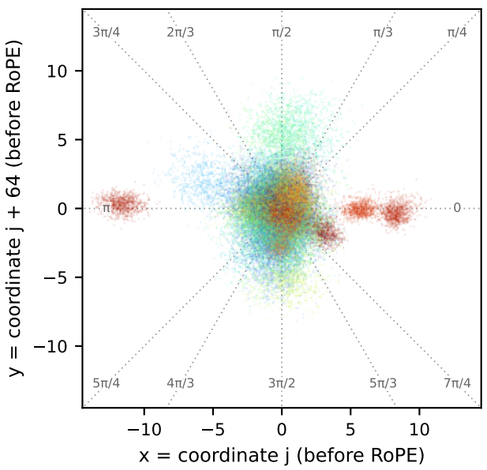
L24 H5



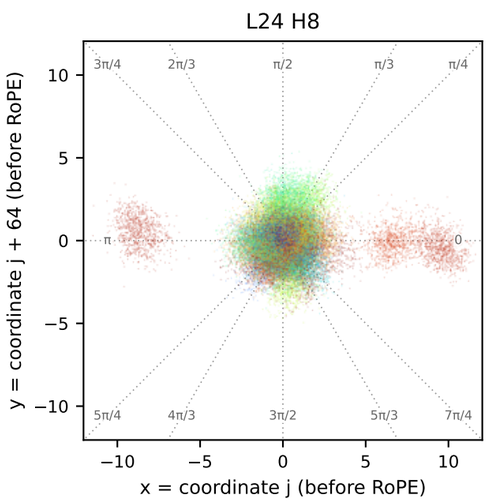
L24 H6



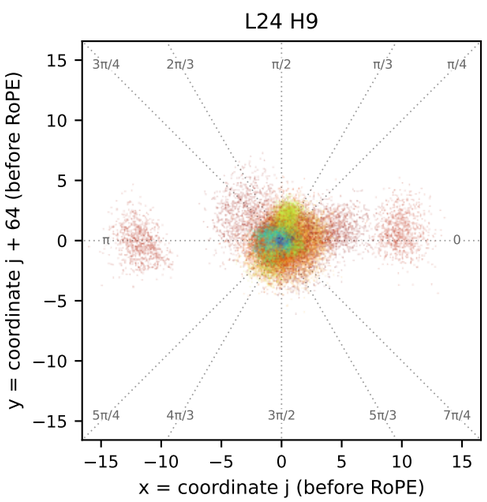
L24 H7



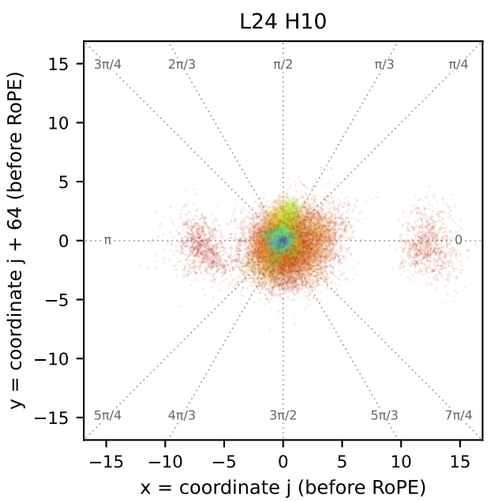
L24 H8



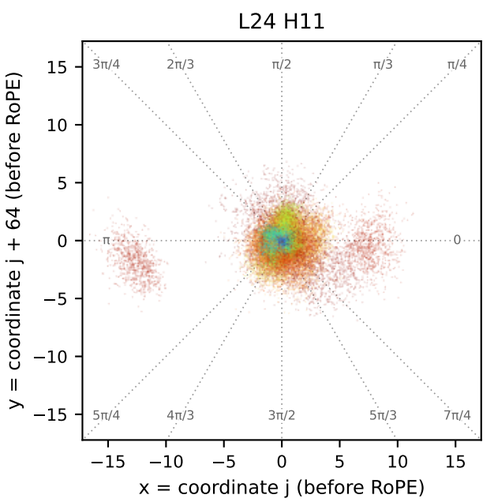
L24 H9

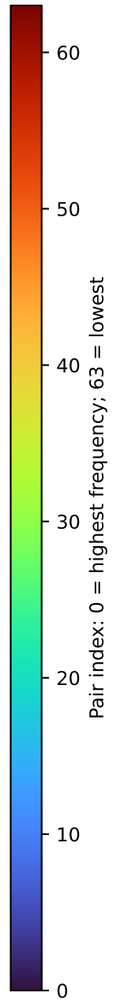


L24 H10



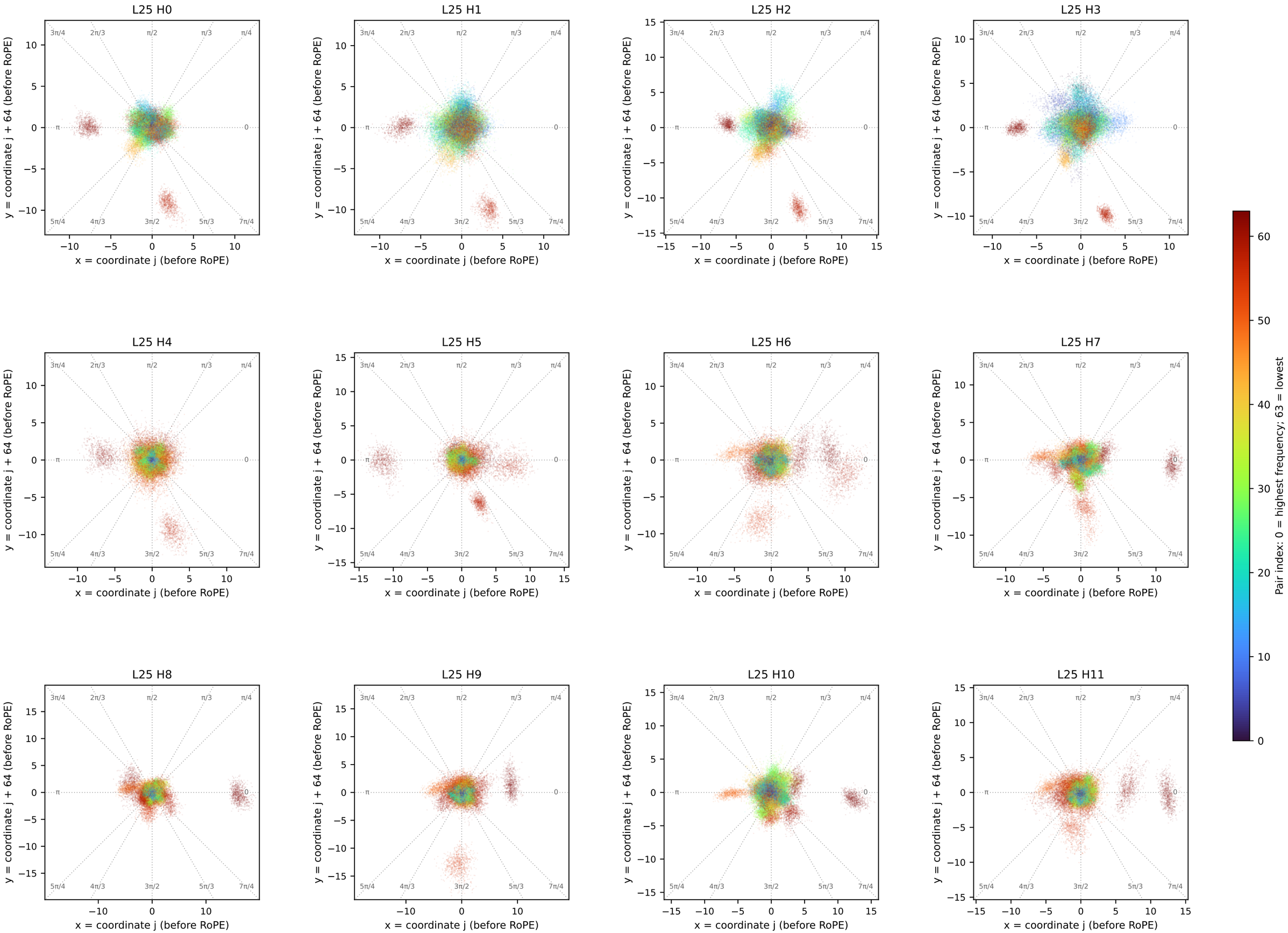
L24 H11





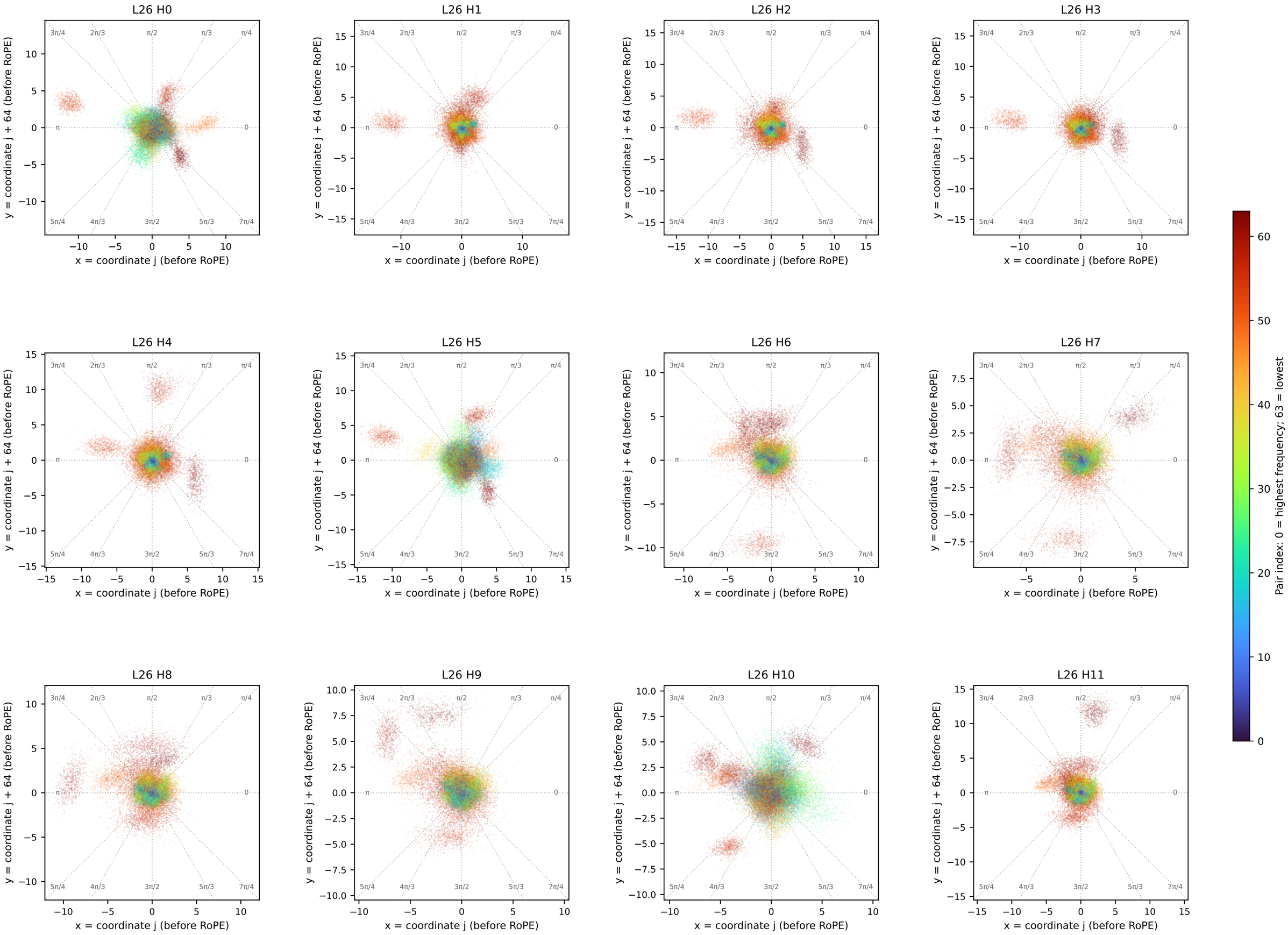
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 25 · axes fitted separately for each head



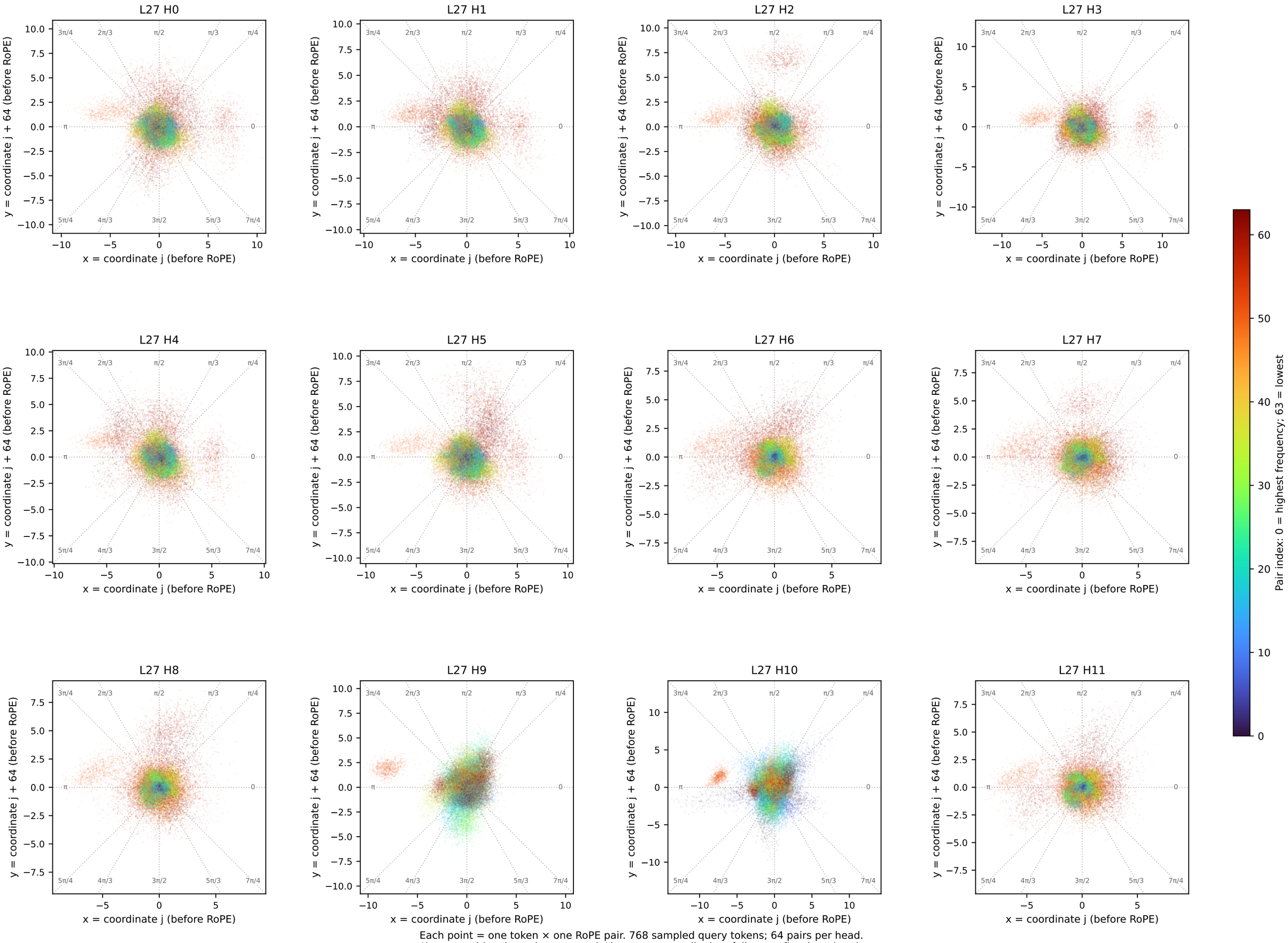
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 26 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 27 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.